

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250279861

Kind Code

A1

Publication Date

September 04, 2025

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MEASUREMENT PROCEDURE

Abstract

Apparatuses and methods for a measurement procedure. A method performed by a user equipment (UE) includes receiving information about a measurement procedure including at least one of at least one measurement resource including a set of time-frequency REs (T-F resource-info), port-related information (port-info), and reference signal (RS)-related information (RS-info). The method further includes receiving a downlink (DL) signal according to the information and, based on the information and the received DL signal, performing the measurement procedure.

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Family ID: 96880634

Appl. No.: 19/056375

Filed: February 18, 2025

Related U.S. Application Data

us-provisional-application US 63560568 20240301

us-provisional-application US 63635964 20240418

us-provisional-application US 63637318 20240422

Publication Classification

Int. Cl.: H04L5/00 (20060101); H04B7/06 (20060101)

U.S. Cl.:

CPC H04L5/0044 (20130101); H04B7/0626 (20130101); H04L5/0005 (20130101);

Background/Summary

CROSS-REFERENCE TO RELATED AND CLAIM OF PRIORITY [0001] The present application claims priority under 35 U.S.C. § 119(e) to U.S. Provisional Patent Application No. 63/560,568 filed on Mar. 1, 2024; U.S. Provisional Patent Application No. 63/635,964 filed on Apr. 18, 2024; and U.S. Provisional Patent Application No. 63/637,318 filed on Apr. 22, 2024, which are hereby incorporated by reference in their entirety.

TECHNICAL FIELD

[0002] The present disclosure relates generally to wireless communication systems and, more specifically, the present disclosure is related to apparatuses and methods for measurement procedure.

BACKGROUND

[0003] Wireless communication has been one of the most successful innovations in modern history. Recently, the number of subscribers to wireless communication services exceeded five billion and continues to grow quickly. The demand of wireless data traffic is rapidly increasing due to the growing popularity among consumers and businesses of smart phones and other mobile data devices, such as tablets, “note pad” computers, net books, eBook readers, and machine type of devices. In order to meet the high growth in mobile data traffic and support new applications and deployments, improvements in radio interface efficiency and coverage are of paramount importance. To meet the demand for wireless data traffic having increased since deployment of 4G communication systems, and to enable various vertical applications, 5G communication systems have been developed and are currently being deployed.

SUMMARY

[0004] The present disclosure relates to measurement procedure.

[0005] In one embodiment, a user equipment (UE) is provided. The UE includes a transceiver configured to receive information about a measurement procedure including at least one of at least one measurement resource including a set of time-frequency REs (T-F resource-info), port-related information (port-info), and reference signal (RS)-related information (RS-info) and receive a downlink (DL) signal according to the information. The UE further includes a processor operably coupled to the transceiver. The processor is configured to, based on the information and the received DL signal, perform the measurement procedure.

[0006] In another embodiment, a base station (BS) is provided. The BS includes a processor and a transceiver operably coupled to the processor. The transceiver is configured to transmit information about a measurement procedure including at least one of at least one measurement resource including a set of T-F resource-info, port-info, and RS-info, and transmit a DL signal according to the information. The measurement procedure is based on the information and the DL signal.

[0007] In yet another embodiment, a method performed by a UE is provided. The method includes receiving information about a measurement procedure including at least one of at least one measurement resource including a set of T-F resource-info, port-info, and RS-info. The method further includes receiving a DL signal according to the information and, based on the information and the received DL signal, performing the measurement procedure.

[0008] Other technical features may be readily apparent to one skilled in the art from the following figures, descriptions, and claims.

[0009] Before undertaking the DETAILED DESCRIPTION below, it may be advantageous to set forth definitions of certain words and phrases used throughout this patent document. The term “couple” and its derivatives refer to any direct or indirect communication between two or more elements, whether or not those elements are in physical contact with one another. The terms “transmit,” “receive,” and “communicate,” as well as derivatives thereof, encompass both direct

and indirect communication. The terms “include” and “comprise,” as well as derivatives thereof, mean inclusion without limitation. The term “or” is inclusive, meaning and/or. The phrase “associated with,” as well as derivatives thereof, means to include, be included within, interconnect with, contain, be contained within, connect to or with, couple to or with, be communicable with, cooperate with, interleave, juxtapose, be proximate to, be bound to or with, have, have a property of, have a relationship to or with, or the like. The term “controller” means any device, system, or part thereof that controls at least one operation. Such a controller may be implemented in hardware or a combination of hardware and software and/or firmware. The functionality associated with any particular controller may be centralized or distributed, whether locally or remotely. The phrase “at least one of,” when used with a list of items, means that different combinations of one or more of the listed items may be used, and only one item in the list may be needed. For example, “at least one of: A, B, and C” includes any of the following combinations: A, B, C, A and B, A and C, B and C, and A and B and C.

[0010] Moreover, various functions described below can be implemented or supported by one or more computer programs, each of which is formed from computer readable program code and embodied in a computer readable medium. The terms “application” and “program” refer to one or more computer programs, software components, sets of instructions, procedures, functions, objects, classes, instances, related data, or a portion thereof adapted for implementation in a suitable computer readable program code. The phrase “computer readable program code” includes any type of computer code, including source code, object code, and executable code. The phrase “computer readable medium” includes any type of medium capable of being accessed by a computer, such as read only memory (ROM), random access memory (RAM), a hard disk drive, a compact disc (CD), a digital video disc (DVD), or any other type of memory. A “non-transitory” computer readable medium excludes wired, wireless, optical, or other communication links that transport transitory electrical or other signals. A non-transitory computer readable medium includes media where data can be permanently stored and media where data can be stored and later overwritten, such as a rewritable optical disc or an erasable memory device.

[0011] Definitions for other certain words and phrases are provided throughout this patent document. Those of ordinary skill in the art should understand that in many if not most instances, such definitions apply to prior as well as future uses of such defined words and phrases.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0012] For a more complete understanding of the present disclosure and its advantages, reference is now made to the following description taken in conjunction with the accompanying drawings, in which like reference numerals represent like parts:

[0013] FIG. 1 illustrates an example wireless network according to embodiments of the present disclosure;

[0014] FIG. 2 illustrates an example gNodeB (gNB) according to embodiments of the present disclosure;

[0015] FIG. 3 illustrates an example UE according to embodiments of the present disclosure;

[0016] FIGS. 4A and 4B illustrate an example of a wireless transmit and receive paths according to embodiments of the present disclosure;

[0017] FIG. 5 illustrates an example of a transmitter structure for beamforming according to embodiments of the present disclosure;

[0018] FIG. 6 illustrates an example of a transmitter structure for physical downlink shared channel (PDSCH) in a subframe according to embodiments of the present disclosure;

[0019] FIG. 7 illustrates an example of a receiver structure for PDSCH in a subframe according to

embodiments of the present disclosure;

[0020] FIG. **8** illustrates an example of a transmitter structure for physical uplink shared channel (PUSCH) in a subframe according to embodiments of the present disclosure;

[0021] FIG. **9** illustrates an example of a receiver structure for a PUSCH in a subframe according to embodiments of the present disclosure;

[0022] FIG. **10** illustrates a diagram of example radio access network (RAN) configurations according to embodiments of the present disclosure;

[0023] FIG. **11** illustrates a diagram of example functional split points/options according to embodiments of the present disclosure;

[0024] FIG. **12** illustrates an example of a fully digital transmitter structure for beamforming according to embodiments of the present disclosure;

[0025] FIG. **13** illustrates a diagram of an example antenna port layout according to embodiments of the present disclosure;

[0026] FIG. **14** illustrates a timeline of example spatial-domain (SD) units and frequency-domain (FD) units according to embodiments of the present disclosure;

[0027] FIG. **15** illustrates a diagram of example port group (PG) hypotheses according to embodiments of the present disclosure;

[0028] FIG. **16** illustrates a diagram of an example transmission configuration indication (TCI) state configuration according to embodiments of the present disclosure;

[0029] FIG. **17** illustrates a diagram of an example TCI state configuration according to embodiments of the present disclosure;

[0030] FIG. **18** illustrates a diagram of an example analog beam-based network according to embodiments of the present disclosure;

[0031] FIG. **19** illustrates examples of a UE moving on a trajectory located in co-located and distributed PGs according to embodiments of the present disclosure;

[0032] FIG. **20** illustrates a flowchart of an example BS procedure for precoding according to embodiments of the present disclosure; and

[0033] FIG. **21** illustrates an example method performed by a UE in a wireless communication system according to embodiments of the present disclosure.

DETAILED DESCRIPTION

[0034] FIGS. **1-21** discussed below, and the various, non-limiting embodiments used to describe the principles of the present disclosure in this patent document are by way of illustration only and should not be construed in any way to limit the scope of the disclosure. Those skilled in the art will understand that the principles of the present disclosure may be implemented in any suitably arranged system or device.

[0035] To meet the demand for wireless data traffic having increased since deployment of 4G communication systems, and to enable various vertical applications, 5G/NR communication systems have been developed and are currently being deployed. The 5G/NR communication system is implemented in higher frequency (mmWave) bands, e.g., 28 GHz or 60 GHz bands, so as to accomplish higher data rates or in lower frequency bands, such as 6 GHz, to enable robust coverage and mobility support. To decrease propagation loss of the radio waves and increase the transmission distance, the beamforming, massive MIMO, full dimensional MIMO (FD-MIMO), array antenna, an analog beam forming, large scale antenna techniques are discussed in 5G/NR communication systems.

[0036] In addition, in 5G/NR communication systems, development for system network improvement is under way based on advanced small cells, cloud radio access networks (RANs), ultra-dense networks, device-to-device (D2D) communication, wireless backhaul, moving network, cooperative communication, coordinated multi-points (CoMP), reception-end interference cancelation and the like.

[0037] In the 5G system, Hybrid frequency shift keying (FSK) and QAM Modulation (FQAM) and

sliding window superposition coding (SWSC) as an advanced coding modulation (ACM), and filter bank multi carrier (FBMC), non-orthogonal multiple access (NOMA), and sparse code multiple access (SCMA) as an advanced access technology have been developed.

[0038] The discussion of 5G systems and frequency bands associated therewith is for reference as certain embodiments of the present disclosure may be implemented in 5G systems. However, the present disclosure is not limited to 5G systems, or the frequency bands associated therewith, and embodiments of the present disclosure may be utilized in connection with any frequency band. For example, aspects of the present disclosure may also be applied to deployment of 5G communication systems, 6G, or even later releases which may use terahertz (THz) bands.

[0039] The following documents and standards descriptions are hereby incorporated by reference into the present disclosure as if fully set forth herein: [REF 1] 3GPP TS 36.211 v17.3.0, “E-UTRA, Physical channels and modulation;” [REF 2] 3GPP TS 36.212 v17.3.0, “E-UTRA, Multiplexing and Channel coding;” [REF 3] 3GPP TS 36.213 v17.3.0, “E-UTRA, Physical Layer Procedures;” [REF 4] 3GPP TS 36.321 v17.3.0, “E-UTRA, Medium Access Control (MAC) protocol specification;” [REF 5] 3GPP TS 36.331 v17.3.0, “E-UTRA, Radio Resource Control (RRC) Protocol Specification;” [REF 6] 3GPP TR 22.891 v1.2.0; [REF 7] 3GPP TS 38.212 v18.0.0, “E-UTRA, NR, Multiplexing and Channel coding;” [REF 8] 3GPP TS 38.214 v18.0.0, “E-UTRA, NR, Physical layer procedures for data;” [REF 9] 3GPP TS 38.211 v18.0.0, “E-UTRA, NR, Physical channels and modulation;” [REF 10] 3GPP TS 38.104 v18.3.0, “E-UTRA, NR, Physical channels and modulation;” [REF 11] O-RAN.WG4.CONF.0-R003-v09.00, “O-RAN Working Group 4 (Fronthaul Working Group) Conformance Test Specification;” and [REF 12] O-RAN.WG4.CUS.0-R003-v13.00, “O-RAN Working Group 4 (Open Fronthaul Interfaces WG)-Control, User and Synchronization Plane Specification.

[0040] FIGS. **1-21** below describe various embodiments implemented in wireless communications systems and with the use of orthogonal frequency division multiplexing (OFDM) or orthogonal frequency division multiple access (OFDMA) communication techniques. The descriptions of FIGS. **1-3** are not meant to imply physical or architectural limitations to how different embodiments may be implemented. Different embodiments of the present disclosure may be implemented in any suitably arranged communications system.

[0041] FIG. **1** illustrates an example wireless network **100** according to embodiments of the present disclosure. The embodiment of the wireless network **100** shown in FIG. **1** is for illustration only. Other embodiments of the wireless network **100** could be used without departing from the scope of the present disclosure.

[0042] As shown in FIG. **1**, the wireless network **100** includes a gNB **101** (e.g., base station, BS), a gNB **102**, and a gNB **103**. The gNB **101** communicates with the gNB **102** and the gNB **103**. The gNB **101** also communicates with at least one network **130**, such as the Internet, a proprietary Internet Protocol (IP) network, or other data network.

[0043] The gNB **102** provides wireless broadband access to the network **130** for a first plurality of user equipments (UEs) within a coverage area **120** of the gNB **102**. The first plurality of UEs includes a UE **111**, which may be located in a small business; a UE **112**, which may be located in an enterprise; a UE **113**, which may be a WiFi hotspot; a UE **114**, which may be located in a first residence; a UE **115**, which may be located in a second residence; and a UE **116**, which may be a mobile device, such as a cell phone, a wireless laptop, a wireless PDA, or the like. The gNB **103** provides wireless broadband access to the network **130** for a second plurality of UEs within a coverage area **125** of the gNB **103**. The second plurality of UEs includes the UE **115** and the UE **116**. In some embodiments, one or more of the gNBs **101-103** may communicate with each other and with the UEs **111-116** using 5G/NR, long term evolution (LTE), long term evolution-advanced (LTE-A), WiMAX, WiFi, or other wireless communication techniques.

[0044] Depending on the network type, the term “base station” or “BS” can refer to any component (or collection of components) configured to provide wireless access to a network, such as transmit

point (TP), transmit-receive point (TRP), an enhanced base station (eNodeB or eNB), a 5G/NR base station (gNB), a macrocell, a femtocell, a WiFi access point (AP), or other wirelessly enabled devices. Base stations may provide wireless access in accordance with one or more wireless communication protocols, e.g., 5G/NR 3.sup.rd generation partnership project (3GPP) NR, long term evolution (LTE), LTE advanced (LTE-A), high speed packet access (HSPA), Wi-Fi 802.11a/b/g/n/ac, etc. For the sake of convenience, the terms “BS” and “TRP” are used interchangeably in this patent document to refer to network infrastructure components that provide wireless access to remote terminals. Also, depending on the network type, the term “user equipment” or “UE” can refer to any component such as “mobile station,” “subscriber station,” “remote terminal,” “wireless terminal,” “receive point,” or “user device.” For the sake of convenience, the terms “user equipment” and “UE” are used in this patent document to refer to remote wireless equipment that wirelessly accesses a BS, whether the UE is a mobile device (such as a mobile telephone or smartphone) or is normally considered a stationary device (such as a desktop computer or vending machine).

[0045] The dotted lines show the approximate extents of the coverage areas **120** and **125**, which are shown as approximately circular for the purposes of illustration and explanation only. It should be clearly understood that the coverage areas associated with gNBs, such as the coverage areas **120** and **125**, may have other shapes, including irregular shapes, depending upon the configuration of the gNBs and variations in the radio environment associated with natural and man-made obstructions.

[0046] As described in more detail below, one or more of the UEs **111-116** include circuitry, programing, or a combination thereof for a measurement procedure. In certain embodiments, one or more of the BSs **101-103** include circuitry, programing, or a combination thereof to support a measurement procedure.

[0047] Although FIG. **1** illustrates one example of a wireless network, various changes may be made to FIG. **1**. For example, the wireless network **100** could include any number of gNBs and any number of UEs in any suitable arrangement. Also, the gNB **101** could communicate directly with any number of UEs and provide those UEs with wireless broadband access to the network **130**. Similarly, each gNB **102-103** could communicate directly with the network **130** and provide UEs with direct wireless broadband access to the network **130**. Further, the gNBs **101**, **102**, and/or **103** could provide access to other or additional external networks, such as external telephone networks or other types of data networks.

[0048] FIG. **2** illustrates an example gNB **102** according to embodiments of the present disclosure. The embodiment of the gNB **102** illustrated in FIG. **2** is for illustration only, and the gNBs **101** and **103** of FIG. **1** could have the same or similar configuration. However, gNBs come in a wide variety of configurations, and FIG. **2** does not limit the scope of the present disclosure to any particular implementation of a gNB.

[0049] As shown in FIG. **2**, the gNB **102** includes multiple antennas **205a-205n**, multiple transceivers **210a-210n**, a controller/processor **225**, a memory **230**, and a backhaul or network interface **235**.

[0050] The transceivers **210a-210n** receive, from the antennas **205a-205n**, incoming radio frequency (RF) signals, such as signals transmitted by UEs in the wireless network **100**. The transceivers **210a-210n** down-convert the incoming RF signals to generate IF or baseband signals. The IF or baseband signals are processed by receive (RX) processing circuitry in the transceivers **210a-210n** and/or controller/processor **225**, which generates processed baseband signals by filtering, decoding, and/or digitizing the baseband or IF signals. The controller/processor **225** may further process the baseband signals.

[0051] Transmit (TX) processing circuitry in the transceivers **210a-210n** and/or controller/processor **225** receives analog or digital data (such as voice data, web data, e-mail, or interactive video game data) from the controller/processor **225**. The TX processing circuitry

encodes, multiplexes, and/or digitizes the outgoing baseband data to generate processed baseband or IF signals. The transceivers **210a-210n** up-converts the baseband or IF signals to RF signals that are transmitted via the antennas **205a-205n**.

[0052] The controller/processor **225** can include one or more processors or other processing devices that control the overall operation of the gNB **102**. For example, the controller/processor **225** could control the reception of uplink (UL) channel signals and the transmission of downlink (DL) channel signals by the transceivers **210a-210n** in accordance with well-known principles. The controller/processor **225** could support additional functions as well, such as more advanced wireless communication functions. For instance, the controller/processor **225** could support beam forming or directional routing operations in which outgoing/incoming signals from/to multiple antennas **205a-205n** are weighted differently to effectively steer the outgoing signals in a desired direction. As another example, the controller/processor **225** could support methods for a measurement procedure. Any of a wide variety of other functions could be supported in the gNB **102** by the controller/processor **225**.

[0053] The controller/processor **225** is also capable of executing programs and other processes resident in the memory **230**, such as processes to support a measurement procedure. The controller/processor **225** can move data into or out of the memory **230** as required by an executing process.

[0054] The controller/processor **225** is also coupled to the backhaul or network interface **235**. The backhaul or network interface **235** allows the gNB **102** to communicate with other devices or systems over a backhaul connection or over a network. The interface **235** could support communications over any suitable wired or wireless connection(s). For example, when the gNB **102** is implemented as part of a cellular communication system (such as one supporting 5G/NR, LTE, or LTE-A), the interface **235** could allow the gNB **102** to communicate with other gNBs over a wired or wireless backhaul connection. When the gNB **102** is implemented as an access point, the interface **235** could allow the gNB **102** to communicate over a wired or wireless local area network or over a wired or wireless connection to a larger network (such as the Internet). The interface **235** includes any suitable structure supporting communications over a wired or wireless connection, such as an Ethernet or transceiver.

[0055] The memory **230** is coupled to the controller/processor **225**. Part of the memory **230** could include a RAM, and another part of the memory **230** could include a Flash memory or other ROM.

[0056] Although FIG. 2 illustrates one example of gNB **102**, various changes may be made to FIG. 2. For example, the gNB **102** could include any number of each component shown in FIG. 2. Also, various components in FIG. 2 could be combined, further subdivided, or omitted and additional components could be added according to particular needs.

[0057] FIG. 3 illustrates an example UE **116** according to embodiments of the present disclosure. The embodiment of the UE **116** illustrated in FIG. 3 is for illustration only, and the UEs **111-115** of FIG. 1 could have the same or similar configuration. However, UEs come in a wide variety of configurations, and FIG. 3 does not limit the scope of the present disclosure to any particular implementation of a UE.

[0058] As shown in FIG. 3, the UE **116** includes antenna(s) **305**, a transceiver(s) **310**, and a microphone **320**. The UE **116** also includes a speaker **330**, a processor **340**, an input/output (I/O) interface (IF) **345**, an input **350**, a display **355**, and a memory **360**. The memory **360** includes an operating system (OS) **361** and one or more applications **362**.

[0059] The transceiver(s) **310** receives from the antenna(s) **305**, an incoming RF signal transmitted by a gNB of the wireless network **100**. The transceiver(s) **310** down-converts the incoming RF signal to generate an intermediate frequency (IF) or baseband signal. The IF or baseband signal is processed by RX processing circuitry in the transceiver(s) **310** and/or processor **340**, which generates a processed baseband signal by filtering, decoding, and/or digitizing the baseband or IF signal. The RX processing circuitry sends the processed baseband signal to the speaker **330** (such

as for voice data) or is processed by the processor **340** (such as for web browsing data).

[0060] TX processing circuitry in the transceiver(s) **310** and/or processor **340** receives analog or digital voice data from the microphone **320** or other outgoing baseband data (such as web data, e-mail, or interactive video game data) from the processor **340**. The TX processing circuitry encodes, multiplexes, and/or digitizes the outgoing baseband data to generate a processed baseband or IF signal. The transceiver(s) **310** up-converts the baseband or IF signal to an RF signal that is transmitted via the antenna(s) **305**.

[0061] The processor **340** can include one or more processors or other processing devices and execute the OS **361** stored in the memory **360** in order to control the overall operation of the UE **116**. For example, the processor **340** could control the reception of DL channel signals and the transmission of uplink (UL) channel signals by the transceiver(s) **310** in accordance with well-known principles. In some embodiments, the processor **340** includes at least one microprocessor or microcontroller.

[0062] The processor **340** is also capable of executing other processes and programs resident in the memory **360**. For example, the processor **340** may execute processes for a measurement procedure as described in embodiments of the present disclosure. The processor **340** can move data into or out of the memory **360** as required by an executing process. In some embodiments, the processor **340** is configured to execute the applications **362** based on the OS **361** or in response to signals received from gNBs or an operator. The processor **340** is also coupled to the I/O interface **345**, which provides the UE **116** with the ability to connect to other devices, such as laptop computers and handheld computers. The I/O interface **345** is the communication path between these accessories and the processor **340**.

[0063] The processor **340** is also coupled to the input **350**, which includes, for example, a touchscreen, keypad, etc., and the display **355**. The operator of the UE **116** can use the input **350** to enter data into the UE **116**. The display **355** may be a liquid crystal display, light emitting diode display, or other display capable of rendering text and/or at least limited graphics, such as from web sites.

[0064] The memory **360** is coupled to the processor **340**. Part of the memory **360** could include a random-access memory (RAM), and another part of the memory **360** could include a Flash memory or other read-only memory (ROM).

[0065] Although FIG. 3 illustrates one example of UE **116**, various changes may be made to FIG. 3. For example, various components in FIG. 3 could be combined, further subdivided, or omitted and additional components could be added according to particular needs. As a particular example, the processor **340** could be divided into multiple processors, such as one or more central processing units (CPUs) and one or more graphics processing units (GPUs). In another example, the transceiver(s) **310** may include any number of transceivers and signal processing chains and may be connected to any number of antennas. Also, while FIG. 3 illustrates the UE **116** configured as a mobile telephone or smartphone, UEs could be configured to operate as other types of mobile or stationary devices.

[0066] FIG. 4A and FIG. 4B illustrate an example of wireless transmit and receive paths **400** and **450**, respectively, according to embodiments of the present disclosure. For example, a transmit path **400** may be described as being implemented in a gNB (such as gNB **102**), while a receive path **450** may be described as being implemented in a UE (such as UE **116**). However, it will be understood that the receive path **450** can be implemented in a gNB and that the transmit path **400** can be implemented in a UE. In some embodiments, the transmit path **400** and/or receive path **450** is configured for a measurement procedure as described in embodiments of the present disclosure.

[0067] As illustrated in FIG. 4A, the transmit path **400** includes a channel coding and modulation block **405**, a serial-to-parallel (S-to-P) block **410**, a size N Inverse Fast Fourier Transform (IFFT) block **415**, a parallel-to-serial (P-to-S) block **420**, an add cyclic prefix block **425**, and an up-converter (UC) **430**. The receive path **450** includes a down-converter (DC) **455**, a remove cyclic

prefix block **460**, a S-to-P block **465**, a size N Fast Fourier Transform (FFT) block **470**, a parallel-to-serial (P-to-S) block **475**, and a channel decoding and demodulation block **480**.

[0068] In the transmit path **400**, the channel coding and modulation block **405** receives a set of information bits, applies coding (such as a low-density parity check (LDPC) coding), and modulates the input bits (such as with Quadrature Phase Shift Keying (QPSK) or Quadrature Amplitude Modulation (QAM)) to generate a sequence of frequency-domain modulation symbols. The serial-to-parallel block **410** converts (such as de-multiplexes) the serial modulated symbols to parallel data in order to generate N parallel symbol streams, where N is the IFFT/FFT size used in the gNB and the UE. The size N IFFT block **415** performs an IFFT operation on the N parallel symbol streams to generate time-domain output signals. The parallel-to-serial block **420** converts (such as multiplexes) the parallel time-domain output symbols from the size N IFFT block **415** in order to generate a serial time-domain signal. The add cyclic prefix block **425** inserts a cyclic prefix to the time-domain signal. The up-converter **430** modulates (such as up-converts) the output of the add cyclic prefix block **425** to a RF frequency for transmission via a wireless channel. The signal may also be filtered at a baseband before conversion to the RF frequency.

[0069] As illustrated in FIG. **4B**, the down-converter **455** down-converts the received signal to a baseband frequency, and the remove cyclic prefix block **460** removes the cyclic prefix to generate a serial time-domain baseband signal. The serial-to-parallel block **465** converts the time-domain baseband signal to parallel time-domain signals. The size N FFT block **470** performs an FFT algorithm to generate N parallel frequency-domain signals. The (P-to-S) block **475** converts the parallel frequency-domain signals to a sequence of modulated data symbols. The channel decoding and demodulation block **480** demodulates and decodes the modulated symbols to recover the original input data stream.

[0070] Each of the gNBs **101-103** may implement a transmit path **400** that is analogous to transmitting in the downlink to UEs **111-116** and may implement a receive path **450** that is analogous to receiving in the uplink from UEs **111-116**. Similarly, each of UEs **111-116** may implement a transmit path **400** for transmitting in the uplink to gNBs **101-103** and may implement a receive path **450** for receiving in the downlink from gNBs **101-103**.

[0071] Each of the components in FIGS. **4A** and **4B** can be implemented using only hardware or using a combination of hardware and software/firmware. As a particular example, at least some of the components in FIGS. **4A** and **4B** may be implemented in software, while other components may be implemented by configurable hardware or a mixture of software and configurable hardware. For instance, the FFT block **470** and the IFFT block **415** may be implemented as configurable software algorithms, where the value of size N may be modified according to the implementation.

[0072] Furthermore, although described as using FFT and IFFT, this is by way of illustration only and should not be construed to limit the scope of the present disclosure. Other types of transforms, such as Discrete Fourier Transform (DFT) and Inverse Discrete Fourier Transform (IDFT) functions, can be used. It will be appreciated that the value of the variable N may be any integer number (such as 1, 2, 3, 4, or the like) for DFT and IDFT functions, while the value of the variable N may be any integer number that is a power of two (such as 1, 2, 4, 8, 16, or the like) for FFT and IFFT functions.

[0073] Although FIGS. **4A** and **4B** illustrate examples of wireless transmit and receive paths **400** and **450**, respectively, various changes may be made to FIGS. **4A** and **4B**. For example, various components in FIGS. **4A** and **4B** can be combined, further subdivided, or omitted and additional components can be added according to particular needs. Also, FIGS. **4A** and **4B** are meant to illustrate examples of the types of transmit and receive paths that can be used in a wireless network. Any other suitable architectures can be used to support wireless communications in a wireless network.

[0074] FIG. **5** illustrates an example of a transmitter structure **500** for beamforming according to

embodiments of the present disclosure. In certain embodiments, one or more of gNB 102 or UE 116 includes the transmitter structure 500. For example, one or more of antenna 205 and its associated systems or antenna 305 and its associated systems can be included in transmitter structure 500. This example is for illustration only and other embodiments can be used without departing from the scope of the present disclosure.

[0075] Accordingly, embodiments of the present disclosure recognize that Rel-14 LTE and Rel-15 NR support up to 32 channel state indication/information CSI reference signal (CSI-RS) antenna ports which enable an eNB or a gNB to be equipped with a large number of antenna elements (such as 64 or 128). A plurality of antenna elements can then be mapped onto one CSI-RS port. For mmWave bands, although a number of antenna elements can be larger for a given form factor, a number of CSI-RS ports, that can correspond to the number of digitally precoded ports, can be limited due to hardware constraints (such as the feasibility to install a large number of analog-to-digital converters (ADCs)/digital-to-analog converters (DACs) at mmWave frequencies) as illustrated in FIG. 5. Then, one CSI-RS port can be mapped onto a large number of antenna elements that can be controlled by a bank of analog phase shifters 501. One CSI-RS port can then correspond to one sub-array which produces a narrow analog beam through analog beamforming 505. This analog beam can be configured to sweep across a wider range of angles 520 by varying the phase shifter bank across symbols or slots/subframes. The number of sub-arrays (equal to the number of RF chains) is the same as the number of CSI-RS ports $N_{\text{sub.CSI-PORT}}$. A digital beamforming unit 510 performs a linear combination across $N_{\text{sub.CSI-PORT}}$ analog beams to further increase a precoding gain. While analog beams are wideband (hence not frequency-selective), digital precoding can be varied across frequency sub-bands or resource blocks. Receiver operation can be conceived analogously.

[0076] Since the transmitter structure 500 of FIG. 5 utilizes multiple analog beams for transmission and reception (wherein one or a small number of analog beams are selected out of a large number, for instance, after a training duration that is occasionally or periodically performed), the term “multi-beam operation” is used to refer to the overall system aspect. This includes, for the purpose of illustration, indicating the assigned DL or UL TX beam (also termed “beam indication”), measuring at least one reference signal for calculating and performing beam reporting (also termed “beam measurement” and “beam reporting”, respectively), and receiving a DL or UL transmission via a selection of a corresponding RX beam. The system of FIG. 5 is also applicable to higher frequency bands such as >52.6 GHz (also termed frequency range 4 or FR4). In this case, the system can employ only analog beams. Due to the O2 absorption loss around 60 GHz frequency (~10 dB additional loss per 100 m distance), a larger number and narrower analog beams (hence a larger number of radiators in the array) are essential to compensate for the additional path loss.

[0077] The present disclosure relates generally to wireless communication systems and, more specifically, to a measurement procedure in next generation of MIMO systems.

[0078] A communication system includes a DownLink (DL) that conveys signals from transmission points such as Base Stations (BSs) or NodeBs to User Equipments (UEs) and an UpLink (UL) that conveys signals from UEs to reception points such as NodeBs. A UE, also commonly referred to as a terminal or a mobile station, may be fixed or mobile and may be a cellular phone, a personal computer device, or an automated device. An eNodeB, which is generally a fixed station, may also be referred to as an access point or other equivalent terminology. For LTE systems, a NodeB is often referred to as an eNodeB.

[0079] In a communication system, such as LTE, DL signals can include data signals conveying information content, control signals conveying DL Control Information (DCI), and Reference Signals (RS) that are also known as pilot signals. An eNodeB transmits data information through a Physical DL Shared CHannel (PDSCH). An eNodeB transmits DCI through a Physical DL Control CHannel (PDCCH) or an Enhanced PDCCH (EPDCCH)—see also REF 3. An eNodeB transmits acknowledgement information in response to data Transport Block (TB) transmission from a UE in

a Physical Hybrid automatic repeat request (ARQ) Indicator CHannel (PHICH). An eNodeB transmits one or more of multiple types of RS including a UE-Common RS (CRS), a Channel State Information RS (CSI-RS), or a DeModulation RS (DMRS). A CRS is transmitted over a DL system BandWidth (BW) and can be used by UEs to obtain a channel estimate to demodulate data or control information or to perform measurements. To reduce CRS overhead, an eNodeB may transmit a CSI-RS with a smaller density in the time and/or frequency domain than a CRS. DMRS can be transmitted only in the BW of a respective PDSCH or EPDCCH and a UE can use the DMRS to demodulate data or control information in a PDSCH or an EPDCCH, respectively. A transmission time interval for DL channels is referred to as a subframe (or slot) and can have, for example, duration of 1 millisecond.

[0080] DL signals also include transmission of a logical channel that carries system control information. A broadcast control channel (BCCH) is mapped to either a transport channel referred to as a Broadcast CHannel (BCH) when it conveys a Master Information Block (MIB) or to a DL Shared CHannel (DL-SCH) when it conveys a System Information Block (SIB)—see also REF3 and REF 5. Most system information is included in different SIBs that are transmitted using DL-SCH. A presence of system information on a DL-SCH in a subframe (or slot) can be indicated by a transmission of a corresponding PDCCH conveying a codeword with a cyclic redundancy check (CRC) scrambled with a special System Information RNTI (SI-RNTI). Alternatively, scheduling information for a SIB transmission can be provided in an earlier SIB and scheduling information for the first SIB (SIB-1) can be provided by the MIB.

[0081] DL resource allocation is performed in a unit of subframe (or slot) and a group of Physical resource blocks (PRBs). A transmission BW includes of frequency resource units referred to as Resource Blocks (RBs). Each RB includes of $N_{\text{sub.sc.sup.RB}}$ sub-carriers, or Resource Elements (REs), such as 12 REs. A unit of one RB over one subframe (or slot) is referred to as a PRB. A UE can be allocated $M_{\text{sub.PDSCH}}$ RBs for a total of $M_{\text{sub.PDSCH}} \cdot N_{\text{sub.sc.sup.RB}}$ REs for the PDSCH transmission BW.

[0082] UL signals can include data signals conveying data information, control signals conveying UL Control Information (UCI), and UL RS. UL RS includes DMRS and Sounding RS (SRS). A UE transmits DMRS only in a BW of a respective PUSCH or PUCCH. An eNodeB can use a DMRS to demodulate data signals or UCI signals. A UE transmits SRS to provide an eNodeB with an UL CSI. A UE transmits data information or UCI through a respective Physical UL Shared CHannel (PUSCH) or a Physical UL Control CHannel (PUCCH). If a UE needs to transmit data information and UCI in a same UL subframe (or slot), it may multiplex both in a PUSCH. UCI includes Hybrid Automatic Repeat reQuest ACKnowledgement (HARQ-ACK) information, indicating correct (ACK) or incorrect (NACK) detection for a data TB in a PDSCH or absence of a PDCCH detection (DTX), Scheduling Request (SR) indicating whether a UE has data in its buffer, Rank Indicator (RI), and Channel State Information (CSI) enabling an eNodeB to perform link adaptation for PDSCH transmissions to a UE. HARQ-ACK information is also transmitted by a UE in response to a detection of a PDCCH/EPDCCH indicating a release of semi-persistently scheduled PDSCH (see also REF 3).

[0083] An UL subframe (or slot) includes two slots. Each slot includes $N_{\text{sub.symb.sup.UL}}$ symbols for transmitting data information, UCI, DMRS, or SRS. A frequency resource unit of an UL system BW is a RB. A UE is allocated $N_{\text{sub.RB}}$ RBs for a total of $N_{\text{sub.RB}} \cdot N_{\text{sub.sc.sup.RB}}$ REs for a transmission BW. For a PUCCH, $N_{\text{sub.RB}}=1$. A last subframe (or slot) symbol can be used to multiplex SRS transmissions from one or more UEs. A number of subframe (or slot) symbols that are available for data/UCI/DMRS transmission is $N_{\text{sub.symb}}=2 \cdot \text{Math}((N_{\text{sub.symb.sup.UL}}-1)-N_{\text{sub.SRS}})$, where $N_{\text{sub.SRS}}=1$ if a last subframe (or slot) symbol is used to transmit SRS and $N_{\text{sub.SRS}}=0$ otherwise.

[0084] FIG. 6 illustrates an example of a transmitter structure 600 for PDSCH in a subframe

according to embodiments of the present disclosure. For example, transmitter structure **600** can be implemented in gNB **102** of FIG. **1**. This example is for illustration only and other embodiments can be used without departing from the scope of the present disclosure.

[0085] As illustrated in FIG. **6**, information bits **610** are encoded by encoder **620**, such as a turbo encoder, and modulated by modulator **630**, for example using Quadrature Phase Shift Keying (QPSK) modulation. A Serial to Parallel (S/P) converter **640** generates M modulation symbols that are subsequently provided to a mapper **650** to be mapped to REs selected by a transmission BW selection unit **655** for an assigned PDSCH transmission BW, unit **660** applies an Inverse Fast Fourier Transform (IFFT), the output is then serialized by a Parallel to Serial (P/S) converter **670** to create a time domain signal, filtering is applied by filter **680**, and a signal transmitted **690**. Additional functionalities, such as data scrambling, cyclic prefix insertion, time windowing, interleaving, and others are well known in the art and are not shown for brevity.

[0086] FIG. **7** illustrates an example of a receiver structure **700** for PDSCH in a subframe according to embodiments of the present disclosure. For example, receiver structure **700** can be implemented by any of the UEs **111-116** of FIG. **1**. This example is for illustration only and other embodiments can be used without departing from the scope of the present disclosure.

[0087] With reference to FIG. **7**, a received signal **710** is filtered by filter **720**, REs **730** for an assigned reception BW are selected by BW selector **735**, unit **740** applies a Fast Fourier Transform (FFT), and an output is serialized by a parallel-to-serial converter **750**. Subsequently, a demodulator **760** coherently demodulates data symbols by applying a channel estimate obtained from a DMRS or a CRS (not shown), and a decoder **770**, such as a turbo decoder, decodes the demodulated data to provide an estimate of the information data bits **780**. Additional functionalities such as time-windowing, cyclic prefix removal, de-scrambling, channel estimation, and de-interleaving are not shown for brevity.

[0088] FIG. **8** illustrates an example of a transmitter structure **800** for PUSCH in a subframe according to embodiments of the present disclosure. For example, transmitter structure **800** can be implemented in gNB **103** of FIG. **1**. This example is for illustration only and other embodiments can be used without departing from the scope of the present disclosure.

[0089] As illustrated in FIG. **8**, information data bits **810** are encoded by encoder **820**, such as a turbo encoder, and modulated by modulator **830**. A Discrete Fourier Transform (DFT) unit **840** applies a DFT on the modulated data bits, REs **850** corresponding to an assigned PUSCH transmission BW are selected by transmission BW selection unit **855**, unit **860** applies an IFFT and, after a cyclic prefix insertion (not shown), filtering is applied by filter **870** and a signal transmitted **880**.

[0090] FIG. **9** illustrates an example of a receiver structure **900** for a PUSCH in a subframe according to embodiments of the present disclosure; For example, receiver structure **900** can be implemented by the UE **116** of FIG. **3**. This example is for illustration only and other embodiments can be used without departing from the scope of the present disclosure.

[0091] As illustrated in FIG. **9**, a received signal **910** is filtered by filter **920**. Subsequently, after a cyclic prefix is removed (not shown), unit **930** applies a FFT, REs **940** corresponding to an assigned PUSCH reception BW are selected by a reception BW selector **945**, unit **950** applies an Inverse DFT (IDFT), a demodulator **960** coherently demodulates data symbols by applying a channel estimate obtained from a DMRS (not shown), a decoder **970**, such as a turbo decoder, decodes the demodulated data to provide an estimate of the information data bits **980**.

[0092] In the 5G system, Hybrid FSK and QAM Modulation (FQAM) and sliding window superposition coding (SWSC) as an advanced coding modulation (ACM), and filter bank multi carrier (FBMC), non-orthogonal multiple access (NOMA), and sparse code multiple access (SCMA) as an advanced access technology have been developed.

[0093] There are two types of frequency range (FR) defined in 3GPP 5G NR specifications. The sub-6 GHz range is called frequency range 1 (FR1) and millimeter wave range is called frequency

range 2 (FR2). An example of the frequency range for FR1 and FR2 is shown herein.

TABLE-US-00001 TABLE A Frequency range designation Corresponding frequency range FR1 450 MHz-600 MHz FR2 24250 MHz-52600 MHz

[0094] For MIMO in FR1, up to 32 CSI-RS antenna ports is supported, and in FR2, up to 8 CSI-RS antenna ports is supported. In next generation cellular standards (e.g., 6G), in addition to FR1 and FR2, new carrier frequency bands can be evaluated, e.g., FR4 (>52.6 GHz), terahertz (>100 GHz) and upper mid-band (10-15 GHz). The number of CSI-RS ports that can be supported for these new bands is likely to be different from FR1 and FR2. In particular, for 10-15 GHz band, the max number of CSI-RS antenna ports is likely to be more than FR1, due to smaller antenna form factors, and feasibility of fully digital beamforming (as in FR1) at these frequencies. For instance, the number of CSI-RS antenna ports can grow up to 128. Besides, the NW (e.g., the network **130**) deployment/topology at these frequencies is also expected to be denser/distributed, for example, antenna ports distributed at multiple (non-co-located, hence geographically separated) TRPs within a cellular region can be the main scenario of interest, due to which the number of CSI-RS antenna ports for MIMO can be even larger (e.g., up to 256).

[0095] A (spatial or digital) precoding/beamforming can be used across these large number of antenna ports in order to achieve MIMO gains. Depending on the carrier frequency, and the feasibility of radio RF/hardware (HW)-related components, the (spatial) precoding/beamforming can be fully digital or hybrid analog-digital. In fully digital beamforming, there can be one-to-one mapping between an antenna port and an antenna element, or a 'static/fixed' virtualization of multiple antenna elements to one antenna port can be used. Each antenna port can be digitally controlled. Hence, a spatial multiplexing across antenna ports is provided.

[0096] In next generation cellular standards (e.g. 6G), in addition to FR1 and FR2, new carrier frequency bands can be evaluated, e.g., FR4 (>52.6 GHz), terahertz (>100 GHz) and upper mid-band (10-15 GHz). The number of CSI-RS ports that can be supported for these new bands is likely to be different from FR1 and FR2. In particular, for 10-15 GHz band, the max number of CSI-RS antenna ports is likely to be more than FR1, due to smaller antenna form factors, and feasibility of fully digital beamforming (as in FR1) at these frequencies. For instance, the number of CSI-RS antenna ports can grow up to 128. Besides, the NW deployment/topology at these frequencies is also expected to be denser/distributed, for example, antenna ports distributed at multiple (non-co-located, hence geographically separated) TRPs within a cellular region can be the main scenario of interest, due to which the number of CSI-RS antenna ports for MIMO can be even larger (e.g. up to 256).

[0097] Likewise, for a cellular system operating in low carrier frequency in general, a sub-1 GHz frequency range (e.g. less than 1 GHz) as an example, supporting large number of CSI-RS antenna ports (e.g. 32) or many antenna elements at a single location or remote radio head (RRH) or TRP is challenging due to a larger antenna form factor size needed taking into account carrier frequency wavelength than a system operating at a higher frequency such as 2 GHz or 4 GHz. At such low frequencies, the maximum number of CSI-RS antenna ports that can be co-located at a site (or RRH or TRP) can be limited, for example to 8. This limits the spectral efficiency of such systems. In particular, the multiuser MIMO (MU-MIMO) spatial multiplexing gains offered due to large number of CSI-RS antenna ports (such as 32) can't be achieved due to the antenna form factor limitation. One plausible way to operate a system with large number of CSI-RS antenna ports at low carrier frequency is to distribute the physical antenna ports to different panels/RRHs/TRPs, which can be non-collocated. The multiple sites or panels/RRHs/TRPs can still be connected to a single (common) base unit forming a single antenna system, hence the signal transmitted/received via multiple distributed RRHs/TRPs can still be processed at a centralized location.

[0098] As described herein, for low (FR1), high (FR2 and beyond), or mid (6-15 GHz) band, the NW topology/architecture is likely to be more and more distributed in future due to reasons explained herein (e.g. use cases, HW requirements, antenna form factors, mobility etc.). In this

disclosure, such a distributed system is referred to as a distributed MIMO (D-MIMO) or multiple TRP (mTRP) system (multiple antenna port groups, which can be non-co-located). The transmission in such a system can be coherent joint transmission (CJT), i.e., a layer can be transmitted across/using multiple TRPs, or non-coherent joint transmission (NCJT). Due to distributed nature of operation, the groups of antenna ports (or TRPs) need to be calibrated/synchronized by compensating for the non-idealities such as time/frequency/phase offsets non-ideal backhaul across TRPs, due to HW impairments, different delay profiles, and Doppler profile (in high-speed scenarios) associated with different TRPs.

[0099] In one example, a TRP or RRH can be functionally equivalent to (hence can be replaced with) or is interchangeable with one of more of the following: an antenna, or an antenna group (multiple antennae), an antenna port, an antenna port group (multiple ports), a CSI-RS resource, multiple CSI-RS resources, a CSI-RS resource set, multiple CSI-RS resource sets, an antenna panel, multiple antenna panels, a Tx-Rx entity, a (analog) beam, a (analog) beam group, a cell, a cell group.

[0100] FIG. 10 illustrates a diagram of example RAN configurations 1000 according to embodiments of the present disclosure. For example, RAN configurations 1000 can be implemented by the BS 102 of FIG. 1. This example is for illustration only and other embodiments can be used without departing from the scope of the present disclosure.

[0101] In an O-RAN, a TRP can be functionally equivalent to (hence can be replaced with) or is interchangeable with one of more of the following: [0102] One RU or O-RU: a logical node that includes a subset of the eNB/gNB functions (e.g. as listed in clause 4.2 split option 7-2x) [0103] More than one RUs or O-RUs [0104] One or more than one RUs or O-RUs

[0105] Two examples are shown in FIG. 10.

[0106] The following are defined in [REF11 and REF12].

TABLE-US-00002 O-CU O-RAN Central Unit-a logical node hosting PDCP, RRC, SDAP and other control functions O-DU O-RAN Distributed Unit: a logical node hosting RLC/MAC/High-PHY layers based on a lower layer functional split. O-DU in addition hosts an M-Plane instance. O-RU O-RAN Radio Unit: a logical node hosting Low-PHY layer and RF processing based on a lower layer functional split. This is similar to 3GPP's "TRP" or "RRH" but more specific in including the Low-PHY layer (FFT/iFFT, PRACH extraction). O-RU in addition hosts M-Plane instance.

[0107] A typical network up to 5G network (NW) can be described in terms of transmit-receive points (TRPs). For a first frequency range (FR1), i.e., <6 GHz, a TRP can comprise one or more antenna ports, and is fully-digital (i.e. each antenna port is driven by a dedicated baseband processing chain); and for a second frequency range 24.25-52.6 GHz (FR2), i.e., for mmWave frequencies, a TRP comprises one of more antenna panels (sub-arrays), each comprising one or two antenna ports that are controlled by analog phase shifters that result in an analog beam (pointing in certain spatial direction). An antenna port in FR1 can also be beamformed (aka virtualization); however, such a beamforming (BF) is generally static (non-adaptive, hence not requiring measurement and reporting). In FR2, due to large propagation loss at mmWave frequencies, each antenna panel requires dynamic/frequent update of the analog BF, which is often based on (analog) beam measurement and reporting.

[0108] A communication between the 5G NW and a user is broadly based on: (A1) NW resources, and (A2) signaling components, where the former corresponds to spatial-domain, frequency-domain, and time-domain (SD, FD, TD) resources allocated to the user for the communication, and the latter corresponds to components that are signaled over the NW resources. The SD resources can be based on a single TRP (sTRP) or multiple TRPs (mTRP), where mTRP can be (B1) co-located at a site/location or (B2) non-co-located/distributed at multiple sites/locations, where the latter corresponds to a distributed SD resource, hence the corresponding communication hypothesis can be (C1) non-coherent joint transmission (NCJT) where a data stream (layer) is transmitted from

one of the mTRPs, or (C2) coherent JT (CJT), where a data stream (layer) can be transmitted from multiple of the mTRPs. The FD resources can comprise a set of PRBs, and the TD resources can comprise one or multiple time slots (i.e., 1 slot= $N_{\text{sub.symb}}$ consecutive symbols).

[0109] The signaling components include signaling associated with (D1) measurement, (D2) channel state information (CSI) report, and (D3) DL reception or UL transmission.

[0110] For (D1), the user measures channel measurement RSs (CMRs) to estimate the channel condition between the sTRP/mTRP and the user. In case of sTRP, the user can measure a set comprising one or multiple DL measurement resources. For mTRP, the measurement resources can be (E1) one resource set comprising one group per TRP, or (E2) one resource set per TRP. The user can also measure the interference based on interference measurement RSs (IMRs). A CMR can correspond to an analog beam, and can be repeated in multiple symbols for determining user's analog beam.

[0111] For (D2), the user, based on the measurement, determines the CSI and reports it to the NW, where the CSI can be (F1) (analog) beam-related CSI, or (F2) (digital) non-beam-related CSI. For (F1), the user determines one or multiple pairs (indicator, metric), where the indicator indicates a CMR and the metric indicates a (beam) quality (e.g. reference signal received power (RSRP), signal-to-interference-plus-noise ratio (SINR)).

[0112] For (F2), a low-resolution (Type-I) CSI and a high-resolution (Type-II) CSI are supported. The Type-I CSI is based on $L=1$ DFT SD vector per layer, requires low feedback overhead and is expected to work reasonably well for single user (SU)-MIMO. For multiuser-(MU)-MIMO transmission, however, high-resolution Type II CSI capturing multiple dominant directions of the channel is essential in order to suppress inter-user interference. The Type-II CSI is based on a weighted linear combination $L>1$ SD DFT vectors where the weights correspond to coefficients. The FD DFT vectors were additionally introduced enhanced Type-II CSI to reduce the CSI feedback overhead by compressing channel coefficients in both SD and FD. A further enhanced Type-II port-selection (PS) CSI was specified to further reduce the CSI overhead by exploiting a reciprocity of angle-and-delay domain between uplink and downlink channels. Expecting NW performs pre-processing with beamformed CSI-RS to concentrate angle-and-delay domain components in few SD and FD basis directions, the user can be configured to select a subset of antenna ports (at a TRP) and one or two FD vectors. Additionally, a NCJT Type-I CSI was supported for up to two TRPs and multiple (sTRP or NCJT) hypotheses. Furthermore, the enhanced Type-II CSI is extended to support CJT Type-II CSI from mTRP and for high/medium user velocities exploiting time-domain correlation or Doppler-domain information, respectively.

[0113] A transmission configuration indication (TCI) framework is shared between (non-beam-related) CSI and beam management (BM). While the complexity of such a TCI framework is justified for CSI acquisition in FR1, it makes BM procedures less efficient in FR2. Furthermore, the BM procedures can be different for different channels due to their different target scenarios. Having different beam indication/update mechanisms increases the complexity, overhead, and latency of BM. Such drawbacks are especially troublesome for high mobility scenarios (such as highway and high-speed train). These drawbacks motivated a streamlined BM framework for beam-based operations and procedures that is common for data and control, and uplink (UL) and downlink (DL) channels. This framework is referred to as a unified TCI (uTCI) framework, firstly introduced for sTRP and now being enhanced for mTRP.

[0114] The uTCI framework supports signaling of a unified TCI state to a user, where the unified TCI state can be a DL-TCI, an UL-TCI or a joint TCI (J-TCI) state, where a DL-TCI state is applied for receiving DL channels/signals, an UL-TCI state is applied for transmitting UL channels/signals, and a J-TCI state is applied for both DL and UL channels/signals. The uTCI framework is designed to support DL receptions and UL transmissions (i) with a joint (common) beam indication for DL and UL by leveraging beam correspondence (reciprocity between DL and UL), and (ii) with separate beam indications for DL and UL, for example to mitigate maximum

permissible exposure, where the beam direction of an UL transmission is different from the beam direction of a DL reception to avoid exposure of the human body to radiation.

[0115] The uTCI framework can support a beam-level mobility, known as inter-cell BM (ICBM). In ICBM, the user-dedicated channels can be configured to use a beam (i.e., TCI state) associated with a (non-serving) cell having a physical cell identity (PCI) that is different from the PCI of the serving cell. This allows fast beam-switch to a non-serving cell for user-dedicated channels at a lower layer without involving a higher layer and without incurring latency and overhead of handover.

[0116] The ICBM is being further enhanced to support a complete cell-switch triggered by lower layers, which is known as lower-layer triggered mobility (LTM). In LTM, the NW can acquire beam measurements, and UL timing information for target candidate cells before cell-switch. The lower layers of the NW decide when to perform a cell-switch, and send a medium access control channel element (MAC CE) containing a cell-switch command (CSC) that triggers the cell-switch from a source cell to a target cell. The CSC includes beam (i.e., TCI state) and UL timing information for the user to use on the target cell. After a beam application delay, the user and the NW communicates via the target cell.

[0117] NW energy saving (NES) is another advanced feature, wherein the NW can optimize energy usage by turning TRPs ON/OFF, thereby saving power. From a user's perspective, this is akin to dynamic SD resource update between transmissions. The CSI in the NES scenario can be based on multiple sub-configurations, each corresponding to different SD resource assignments, and dynamically (via DCI) triggering one or multiple of the sub-configurations for CSI reporting.

[0118] Full-duplex transmission and reception in the same NR channel BW or using non-contiguous intra-band carrier aggregation (CA) is a promising technology to enhance UL coverage, reduce latency and improve system capacity and to overcome limitations inherent to the use of de-facto mandated semi-static time division duplexing (TDD) UL-DL frame configurations in today's NR TDD deployments. Currently, 3GPP is studying benefits, feasibility and deployment scenarios for enabling NW-side full-duplex operation where simultaneous transmissions and receptions by the NW on the same time-domain symbol on the NR carrier can only occur in non-overlapping UL and DL subbands, e.g., subband full-duplex (SBFD) mode. In this first step of NR duplex evolution, users with support for NW-side SBFD operation still operate in half-duplex, i.e., the user can either transmit or receive on an SBFD symbol but not transmit and receive simultaneously. An SBFD UL subband can be located in the center or at the edge of the NR carrier in FR1 or FR2-1. For CA-based SBFD in FR2-1, one component carrier (CC) is allocated for UL transmissions whereas the remaining CCs are used for DL transmission.

[0119] NW-side self-interference cancellation (SIC) capability to enable SBFD can be realized through a combination of solutions. For example, the NW can use Tx/Rx antenna isolation on the antenna panel(s), beam steering, analog and/or digital pre-distortion, digital interference cancellation, and analog and/or digital filtering solutions. Note that passive Tx/Rx antenna isolation has been demonstrated to achieve in excess of 80 dB in FR1 with even higher isolation in FR2-1. For example, SBFD for the Local Area base station class characterized by small Tx power and reduced Rx sensitivity can already achieve a significant amount of SIC capability by relying on antenna isolation alone. Wide Area base stations characterized by much higher transmit power and higher Rx sensitivity may need to implement a more extensive set of solutions to support SBFD.

[0120] NW-side SBFD operation can be enabled transparently for typical NR users and has been shown feasible and providing gains. In this case, typical users are scheduled UL transmissions in the SBFD UL subband of the NR carrier on symbols configured as flexible by SIB1. More gains in the NR TDD cell supporting NW-side SBFD operation can be achieved in presence of SBFD-aware users, e.g., supporting resource allocation enhancements for PDSCH, PUSCH and PUCCH including handling of TCI states and BF, and CSI reporting enhancements to best exploit the link conditions on SBFD and non-SBFD slots/symbols on the serving cell.

[0121] As explained, the 5G NW can support several features, services, use cases, and deployment scenarios. It however also introduces too many different abstractions (for specification) of NW entities and involved signaling for components of these abstractions. For instance, the specification supports abstractions for single-cell, multi-cell, sTRP, mTRP, panel, antenna panel, antenna port, resource, resource set, and beam; and signaling for a complicated CSI framework based on components such as CSI resource setting (one or more CSI-RS resources sets, each with one or more CSI-RS/synchronization signal block (SSB) resources) and CSI report setting that links a CSI resource setting to a report quantity from a set of multiple supported report quantities, wherein a report quantity can correspond to beam report (i.e. an analog beam and a beam quality) or a non-beam report (i.e. rank indicator (RI), precoding matrix indicator (PMI), channel quality indicator (CQI), CQI report interval (CRI)). In addition, for PMI, too many different codebooks are supported. Due to these reasons, deployment of the 5G NW is challenging in reality. A direct scaling/extension/reuse of these typical up to 5G solutions for 6G will add to the complexity, which is undesired in real NW deployments.

[0122] FIG. 11 illustrates a diagram of example functional split points/options 1100 according to embodiments of the present disclosure. For example, functional split points/options 1100 may be implemented by the BS 102 of FIG. 1. This example is for illustration only and other embodiments can be used without departing from the scope of the present disclosure.

[0123] In next-gen MIMO systems (e.g. 6G), at least two aspects need to be evaluated. [0124] (A) 3GPP PHY specification: The significance of a single NW entity, namely PG (as a collection of ports) in terms of port-common channel properties. This is analogous to the 5G QCL (or TCI state), coherency assumption (e.g. FC, PC, NC). [0125] (B) NW architecture as perceived in O-RAN: The functionality split among O-RAN entities for DL and UL operations, such as O-RU, O-DU, and O-CU (as described herein). An example is shown in FIG. 11. In particular, the PHY functionality split between O-DU and O-RU includes at least the following aspects. [0126] (B1) PHY processing: [0127] bit-level processing, [0128] symbol-level processing [0129] (B2) Scheduling (residing in MAC): SU-MIMO/MU-MIMO scheduling across different O-RUs or/and allocated frequency-domain resources (e.g. PRBs, precoding resource block groups (PRGs), subbands (SBs)) [0130] Utilizing UCI carrying CSI [0131] If DL/UL reciprocity is feasible, also utilizing SRS-based channel measurement [0132] B3) Precoder calculation at a gNB (e.g., the BS 102) (NW side) for DL-SCH transmission: [0133] For SU-MIMO, precoder can simply follow the PMI (calculated assuming SU-MIMO hypothesis) reported by the UE, or, if DL/UL reciprocity is feasible, be calculated from the eigenvector(s) of the measured DL channels. [0134] For MU-MIMO, precoder needs to be calculated based on additional orthogonalization (e.g. zero-forcing beamforming (ZFBF), signal-to-leakage-and-noise-ratio (SLNR)) among PMIs, or, if DL/UL reciprocity is feasible, the eigenvectors of the measured channels of the co-scheduled UEs

[0135] The first (A) can be achieved by removing/merging duplicate/redundant abstractions, and simplifying signaling for components of the abstractions. One such framework, namely dynamic MIMO, is provided in this disclosure, wherein abstractions such as CSI-RS resource, CSI-RS resource set, port, beam, TRP, panel etc. can be clubbed into one basic entity, namely antenna/port group (PG or O-RU (or RU)), and essential features of PGs are specified. A few essential features discussed include dynamic PG or O-RU (or RU) selection and long-term stats and assumptions across PGs, e.g., quasi co-location (QCL) and coherency relationships across PGs. The provided framework can also facilitate fast and accurate CSI acquisition, where the CSI can be beam-related (e.g. beam indicator, beam metric), non-beam-related (e.g. RI/PMI/CQI), or both. Additionally, the concept of a cell is replaced with PGs that are distributed through the NW. The mobility can be handled via the PG or O-RU (or RU) selection/update (from one set of PGs to another set of PGs). [0136] A few relevant (more-probable) candidates discussed in the O-RAN Alliance (depicted in FIG. 11) are shown in Table 0.

TABLE-US-00003 TABLE 0 (both DL and UL) High- Low- PDCP RLC MAC PHY PHY RF HLS

LLS O-MAC-RAN.sup.1 O-CU: O-DU: RLC, MAC, O-RU: Low-PHY, Y symbol- (Opt7-2x) PDCP High-PHY RF level PHY Opt7-3 O-CU: O-DU: RLC, MAC, O-RU: Low-PHY, Y bit-level PDCP High-PHY RF PHY Opt8 DU: RLC, MAC, PHY RU: Y CPRI RF O-RAN.sup.1: [REF 12] [0137] Cat-A, Cat-B [0138] UL: Cat-C

[0139] The 5G NR MIMO inherits a number of unnecessary hierarchical specification entities from 4G LTE. In relation to multi-antenna (MIMO), such entities include: [0140] Pointless abstractions: resource and resource-set entities for RS [0141] Obsolete implementation-based abstractions: panel, multi-panel, “TRP”, FR1 port vs FR2 beam/resource, “cell”

[0142] Going into 6G, the MIMO framework should be simplified/streamlined in order to (i) support both typical (up to 5G) and new frequency bands (e.g. FR3), (ii) enable new features/services (such as AI/ML-based learning, evolved duplexing, and energy saving), (iii) make it implementation friendly, and (iv) have a future-proof and easily upgradable system.

[0143] FIG. 12 illustrates an example of a fully digital transmitter structure 1200 for beamforming according to embodiments of the present disclosure. For example, fully digital transmitter structure 1200 can be implemented in the BS 102 of FIG. 2. This example is for illustration only and other embodiments can be used without departing from the scope of the present disclosure.

[0144] The fully digital transmitter structure 1200 includes a digital beamformer 1210, a fixed beam/virtualization 1201, and antenna array angles 1220.

[0145] A 5G NW can be built upon a spatial resource entity, say X. For a first frequency range (FR1), i.e., <6 GHz, the spatial entity X comprises one or more antenna ports that are fully-digital (i.e. each antenna port is driven by a dedicated baseband processing chain), as shown in FIG. 12; and for a second frequency range 24.25-52.6 GHz (FR2), i.e., for mmWave frequencies, the entity X comprises one or more antenna panels (sub-arrays), each comprising one (or two) antenna ports that is (are) controlled by analog phase shifters that result in an analog beam (pointing in certain spatial direction), as shown in FIG. 5. An antenna port in FR1 can also be beam-formed (aka virtualization); however, such a beamforming (BF) is generally static (non-adaptive, hence does not requiring measurement and reporting). In FR2, due to large propagation loss at mmWave frequencies, each antenna panel requires dynamic/frequent update of the analog BF, which is often based on (analog) beam measurement and reporting.

[0146] Thus, the main difference between a FR1 port and a FR2 panel is that the beam/virtualization (i.e. port assignment) is fixed in the former, and it requires (a) measurement and reporting from UE (e.g., the UE 116) and (b) a beam indication from the NW (TCI state with QCL-TypeD). It is therefore plausible to have a unified framework in which a port in FR1 and a panel in FR2 can be abstracted based on a unified, band-agnostic spatial entity, e.g. port or port group (PG), and associated QCL and coherency properties across ports or PGs (intra-/inter PG). For instance, [0147] Port: a FR1 port and a FR2 beam/source [0148] Port group (PG): TRP, resource, resource set, panel, cell

[0149] While the O-RAN Alliance is intended for 5G NR, it is expected that its framework will continue, or at most refined, for 6G. The O-RAN Alliance specifies 3 levels of functional splits—namely CU, DU, and RU—to facilitate multi-vendor inter-operability within a NW. The manner in which PHY-layer functions are split between DU and RU(s) imposes serious impact on the feasibility, performance, and complexity of different MIMO schemes—mainly due to the latency and quantization loss incurred by the O-RAN-standardized RU-DU interface.

[0150] Instead of reusing the 4G/5G abstraction of CSI-RS resource or resource set, the terms “port” as a spatial-domain resource unit and “port group” (PG) as a collection of ports sharing a same set of channel properties are used irrespective of the frequency band. In this sense, a “port” can be associated with a digital port in FR1 or an analog beam in FR2 (thereby abandoning the 5G association between an analog beam and a CSI-RS resource for FR2).

[0151] The present disclosure relates to next generation communication systems (e.g. adv. 5G and 6G) based on a dynamic MIMO framework wherein there is a single/basis abstraction entity,

namely antenna/port group (PG) that is used over wide range of applications such as common MIMO (FR1), beam-formed MIMO (FR2 and beyond), MIMO in mid-band (e.g. 7-15 GHz, FR3), dynamic port/PG assignment/update for network energy saving, duplexing operations, predictive MIMO (mobility scenarios) etc. This disclosure, in particular provides, a notion of measurement region based on three dimensions (SD, FD, TD), and following are aspects of this disclosure: [0152] Spatial-domain: one or multiple PGs [0153] N.sub.g: number of ports comprising a PG [0154] QCL-relation/assumptions among ports with a PG [0155] QCL-relation/assumptions across ports in multiple PGs [0156] Time-domain/frequency-domain measurement: a set of REs $T=\{(k, l)\}$, k is an RE index and l is a symbol index [0157] Mapping/association of ports (within N.sub.g PGs) to the set of REs T [0158] Power level: fixed vs adaptive power [0159] Signaling details: number of ports, mapping ξ [0160] Higher layer [0161] MAC CE [0162] Dynamic (DCI) [0163] Aspects, features, and advantages of the disclosure are readily apparent from the following detailed description, simply by illustrating a number of particular embodiments and implementations, including the best mode contemplated for carrying out the disclosure. The disclosure is also capable of other and different embodiments, and its several details can be modified in various obvious respects, all without departing from the spirit and scope of the disclosure. Accordingly, the drawings and description are to be regarded as illustrative in nature, and not as restrictive. The disclosure is illustrated by way of example, and not by way of limitation, in the figures of the accompanying drawings.

[0164] In the following, for brevity, both frequency division duplexing (FDD) and time division duplexing (TDD) are provided as the duplex method for both DL and UL signaling.

[0165] Although exemplary descriptions and embodiments to follow expect orthogonal frequency division multiplexing (OFDM) or orthogonal frequency division multiple access (OFDMA), this disclosure can be extended to other OFDM-based transmission waveforms or multiple access schemes such as filtered OFDM (F-OFDM).

[0166] This disclosure of disclosure covers several components which can be used in conjunction or in combination with one another, or can operate as standalone schemes.

[0167] All the following components and embodiments are applicable for UL transmission with CP-OFDM (cyclic prefix OFDM) waveform as well as DFT-SOFDM (DFT-spread OFDM) and SC-FDMA (single-carrier FDMA) waveforms. Furthermore, the following components and embodiments are applicable for UL transmission when the scheduling unit in time is either one subframe (which can include one or multiple slots) or one slot.

[0168] In the present disclosure, the frequency resolution (reporting granularity) and span (reporting bandwidth) of CSI reporting can be defined in terms of frequency “subbands” and “CSI reporting band” (CRB), respectively.

[0169] A subband for CSI reporting is defined as a set of contiguous PRBs which represents the smallest frequency unit for CSI reporting. The number of PRBs in a subband can be fixed for a given value of DL system bandwidth, configured either semi-statically via higher-layer/RRC signaling, or dynamically via L1 DL control signaling or MAC control element (MAC CE). The number of PRBs in a subband can be included in CSI reporting setting.

[0170] “CSI reporting band” is defined as a set/collection of subbands, either contiguous or non-contiguous, wherein CSI reporting is performed. For example, CSI reporting band can include the subbands within the DL system bandwidth. This can also be termed “full-band”. Alternatively, CSI reporting band can include only a collection of subbands within the DL system bandwidth. This can also be termed “partial band”.

[0171] The term “CSI reporting band” is used only as an example for representing a function. Other terms such as “CSI reporting subband set” or “CSI reporting bandwidth” or bandwidth part (BWP) can also be used.

[0172] In terms of UE configuration, a UE can be configured with at least one CSI reporting band. This configuration can be semi-static (via higher-layer signaling or RRC) or dynamic (via MAC

CE or L1 DL control signaling). When configured with multiple (N) CSI reporting bands (e.g. via RRC signaling), a UE can report CSI associated with $n \leq N$ CSI reporting bands. For instance, >6 GHz, large system bandwidth may require multiple CSI reporting bands. The value of n can either be configured semi-statically (via higher-layer signaling or RRC) or dynamically (via MAC CE or L1 DL control signaling). Alternatively, the UE can report a recommended value of n via an UL channel.

[0173] Therefore, CSI parameter frequency granularity can be defined per CSI reporting band as follows. A CSI parameter is configured with “single” reporting for the CSI reporting band with $M_{\text{sub},n}$ subbands when one CSI parameter for the $M_{\text{sub},n}$ subbands within the CSI reporting band. A CSI parameter is configured with “subband” for the CSI reporting band with $M_{\text{sub},n}$ subbands when one CSI parameter is reported for each of the $M_{\text{sub},n}$ subbands within the CSI reporting band.

[0174] FIG. 13 illustrates a diagram of an example antenna port layout 1300 according to embodiments of the present disclosure. For example, antenna port layout 1300 can be implemented in the wireless network 100 of FIG. 1. This example is for illustration only and can be used without departing from the scope of the present disclosure.

[0175] In the following, $N_{\text{sub},1}$ and $N_{\text{sub},2}$ are the number of antenna ports with the same polarization in the first and second dimensions, respectively. For 2D antenna port layouts, $N_{\text{sub},1} > 1$, $N_{\text{sub},2} > 1$, and for 1D antenna port layouts $N_{\text{sub},1} > 1$ and $N_{\text{sub},2} = 1$ or $N_{\text{sub},2} > 1$ and $N_{\text{sub},1} = 1$. In the rest of the disclosure, 1D antenna port layouts with $N_{\text{sub},1} > 1$ and $N_{\text{sub},2} = 1$ is provided. The disclosure, however, is applicable to the other 1D port layouts with $N_{\text{sub},2} > 1$ and $N_{\text{sub},1} = 1$. Also, in the rest of the disclosure, $N_{\text{sub},1} \geq N_{\text{sub},2}$. The disclosure, however, is applicable to the case when $N_{\text{sub},1} < N_{\text{sub},2}$, and the embodiments for $N_{\text{sub},1} > N_{\text{sub},2}$ apply to the case $N_{\text{sub},1} < N_{\text{sub},2}$ by swapping/switching $(N_{\text{sub},1}, N_{\text{sub},2})$ with $(N_{\text{sub},2}, N_{\text{sub},1})$. For a single-polarized (or co-polarized) antenna port layout, the total number of antenna ports is $P_{\text{sub},\text{CSIRS}} = N_{\text{sub},1}N_{\text{sub},2}$. And, for a dual-polarized antenna port layout, the total number of antenna ports is $P_{\text{sub},\text{CSIRS}} = 2N_{\text{sub},1}N_{\text{sub},2}$. An illustration is shown in FIG. 13 where “X” represents two antenna polarizations (dual-pol, $s=2$) and “/” represents one antenna polarization (co-pol, $s=1$). In this disclosure, the term “polarization” refers to a group of antenna ports with the same polarization. For example, antenna ports $j = X+0, X+1, \dots$,

$$[00001] X + \frac{P_{\text{sub},\text{CSIRS}}}{2} - 1$$

comprise a first antenna polarization, and antenna ports

$$[00002] j = X + \frac{P_{\text{sub},\text{CSIRS}}}{2}, X + \frac{P_{\text{sub},\text{CSIRS}}}{2} + 1,$$

$\dots, X + P_{\text{sub},\text{CSIRS}} - 1$ comprise a second antenna polarization, where $P_{\text{sub},\text{CSIRS}}$ is a number of CSI-RS antenna ports and X is a starting antenna port number (e.g. $X=3000$, then antenna ports are 3000, 3001, 3002, $\dots$). Unless stated otherwise, dual-polarized antenna layouts are expected in this disclosure. The embodiments (and examples) in this disclosure however are general and are applicable to single-polarized antenna layouts as well.

[0176] Let s denotes the number of antenna polarizations (or groups of antenna ports with the same polarization). Then, for co-polarized antenna ports, $s=1$, and for dual- or cross (X)-polarized antenna ports $s=2$. So, the total number of antenna ports $P_{\text{sub},\text{CSIRS}} = sN_{\text{sub},1}N_{\text{sub},2}$

[0177] Let $N_{\text{sub},g}$ be a number of antenna/port groups (PGs). When there are multiple antenna/port groups ($N_{\text{sub},g} > 1$), each group ($g \in \{1, \dots, N_{\text{sub},g}\}$) comprises $N_{\text{sub},1,g}$ and $N_{\text{sub},2,g}$ ports in two dimensions. This is illustrated in FIG. 13. Note that the antenna port layouts may be the same ($N_{\text{sub},1,g} = N_{\text{sub},1}$ and $N_{\text{sub},2,g} = N_{\text{sub},2}$) in different antenna/port groups, or they can be different across antenna/port groups. For group g , the number of antenna ports is $P_{\text{sub},\text{CSIRS},g} = N_{\text{sub},1,g}N_{\text{sub},2,g}$ or $2N_{\text{sub},1,g}N_{\text{sub},2,g}$ (for co-polarized or dual-polarized respectively), i.e., $P_{\text{sub},\text{CSIRS},g} = s_{\text{sub},g}N_{\text{sub},1,g}N_{\text{sub},2,g}$ where $s_{\text{sub},g} = 1$ or 2

[0178] In one example, an antenna/port group corresponds to an antenna panel. In one example, an antenna/port group corresponds to a TRP. In one example, an antenna/port group corresponds to an

RRH. In one example, an antenna/port group corresponds to CSI-RS antenna ports of a non zero power (NZP) CSI-RS resource. In one example, an antenna/port group corresponds to a subset of CSI-RS antenna ports of a NZP CSI-RS resource (comprising multiple antenna/port groups). In one example, an antenna/port group corresponds to CSI-RS antenna ports of multiple NZP CSI-RS resources (e.g. comprising a CSI-RS resource set).

[0179] In one example, an antenna/port group corresponds to a reconfigurable intelligent surface (RIS) in which the antenna/port group can be (re-)configured more dynamically (e.g. via MAC CE or/and DCI). For example, the number of antenna ports associated with the antenna/port group can be changed dynamically.

[0180] In one example, the antenna architecture of the MIMO system is structured. For example, the antenna structure at each PG or O-RU (or RU) is dual-polarized (single or multi-panel as shown in FIG. 13). The antenna structure at each PG or O-RU (or RU) can be the same. Or, the antenna structure at an PG or O-RU (or RU) can be different from another PG or O-RU (or RU). Likewise, the number of ports at each PG (or O-RU or RU) can be the same. Or, the number of ports at one PG (or O-RU or RU) can be different from another PG (or O-RU or RU).

[0181] In another example, the antenna architecture of the MIMO system is unstructured. For example, the antenna structure at one PG (or O-RU or RU) can be different from another PG (or O-RU or RU).

[0182] A structured antenna architecture is expected in the rest of the disclosure. For simplicity, each PG (or O-RU or RU) is equivalent to a panel (cf. FIG. 13), although, an PG (or O-RU or RU) can have multiple panels in practice. The disclosure however is not restrictive to a single panel assumption at each PG (or O-RU or RU), and can easily be extended (covers) the case when an PG (or O-RU or RU) has multiple antenna panels.

[0183] In one embodiment, an PG (or O-RU or RU) constitutes (or corresponds to or is equivalent to) at least one of the following: [0184] In one example, an PG or O-RU (or RU) corresponds to a TRP. [0185] In one example, an PG or O-RU (or RU) corresponds to a CSI-RS resource. A UE is configured with $K=N_{\text{sub}} \cdot g > 1$ non-zero-power (NZP) CSI-RS resources, and a CSI reporting is configured to be across multiple CSI-RS resources. This is similar to Class B, $K > 1$ configuration in Rel. 14 LTE. The K NZP CSI-RS resources can belong to a CSI-RS resource set or multiple CSI-RS resource sets (e.g. K resource sets each comprising one CSI-RS resource). The details are as explained herein in this disclosure. [0186] In one example, an PG or O-RU (or RU) corresponds to a CSI-RS resource group, where a group comprises one or multiple NZP CSI-RS resources. A UE is configured with $K \geq N_{\text{sub}} \cdot g > 1$ non-zero-power (NZP) CSI-RS resources, and a CSI reporting is configured to be across multiple CSI-RS resources from resource groups. This is similar to Class B, $K > 1$ configuration in Rel. 14 LTE. The K NZP CSI-RS resources can belong to a CSI-RS resource set or multiple CSI-RS resource sets (e.g. K resource sets each comprising one CSI-RS resource). The details are as explained herein in this disclosure. In particular, the K CSI-RS resources can be partitioned into $N_{\text{sub}} \cdot g$ resource groups. The information about the resource grouping can be provided together with the CSI-RS resource setting/configuration, or with the CSI reporting setting/configuration, or with the CSI-RS resource configuration. [0187] In one example, an PG or O-RU (or RU) corresponds to a subset (or a group) of CSI-RS ports. A UE is configured with at least one NZP CSI-RS resource comprising (or associated with) CSI-RS ports that can be grouped (or partitioned) multiple subsets/groups/parts of antenna ports, each corresponding to (or constituting) an PG or O-RU (or RU). The information about the subsets of ports or grouping of ports can be provided together with the CSI-RS resource setting/configuration, or with the CSI reporting setting/configuration, or with the CSI-RS resource configuration. [0188] In one example, an PG or O-RU (or RU) corresponds to one or more examples described herein depending on a configuration. For example, this configuration can be explicit via a parameter (e.g. an RRC parameter). Or, it can be implicit. [0189] In one example, when implicit, it could be based on the value of K . For example, when $K > 1$ CSI-RS resources, an PG or O-RU (or RU) corresponds to one

or more examples described herein, and when $K=1$ CSI-RS resource, an PG or O-RU (or RU) corresponds to one or more examples described herein. [0190] In another example, the configuration could be based on the configured codebook. For example, an PG or O-RU (or RU) corresponds to a CSI-RS resource (according to one or more examples described herein) or resource group (according to one or more examples described herein) when the codebook corresponds to a decoupled codebook (modular or separate codebook for each PG or O-RU (or RU)), and an PG or O-RU (or RU) corresponds to a subset (or a group) of CSI-RS ports (according to one or more examples described herein) when codebook corresponds to a coupled (joint or coherent) codebook (one joint codebook across PGs).

[0191] In one example, when PG or O-RU (or RU) maps (or corresponds to) a CSI-RS resource or resource group (according to one or more examples described herein), and a UE can select a subset of PGs (resources or resource groups) and report the CSI for the selected PGs (resources or resource groups), the selected PGs can be reported via an indicator. For example, the indicator can be a channel quality indicator report interval (CRI) or a PMI (component) or a new indicator.

[0192] In one example, when PG or O-RU (or RU) maps (or corresponds to) a CSI-RS port group (according to one or more examples described herein), and a UE can select a subset of PGs (port groups) and report the CSI for the selected PGs (port groups), the selected PGs can be reported via an indicator. For example, the indicator can be a CRI or a PMI (component) or a new indicator.

[0193] In one example, when multiple ($K>1$) CSI-RS resources are configured for N.sub.g PGs (according to one or more examples described herein), a decoupled (modular) codebook is used/configured, and when a single ($K=1$) CSI-RS resource for N.sub.g PGs (according to one or more examples described herein), a joint codebook is used/configured.

[0194] FIG. **14** illustrates a timeline **1400** of example SD units and FD units according to embodiments of the present disclosure. For example, timeline **1400** can be followed by any of the UEs **111-116** of FIG. **1**, such as the UE **116**. This example is for illustration only and other embodiments can be used without departing from the scope of the present disclosure.

[0195] In one embodiment, a UE is configured (e.g. via a higher layer CSI configuration information) with a CSI report, where the CSI report is based on a channel measurement (and interference measurement) and a codebook. When the CSI report is configured to be aperiodic, it is reported when triggered via a DCI field (e.g. a CSI request field) in a DCI.

[0196] The channel measurement can be based on $K \geq 1$ channel measurement reference signals (RSs) or resources (CMRs) that are transmitted from a plurality of spatial-domain (SD) units (e.g. a SD unit=a CSI-RS antenna port), and are measured via a plurality of frequency-domain (FD) units (e.g. a FD unit=one or more PRBs/SBs) and via either a time-domain (TD) unit or a plurality of TD units (e.g. a TD unit=one or more time slots). In one example, a CMR can be a NZP-CSI-RS (or resource).

[0197] The CSI report can be associated with the plurality of FD units and the plurality of TD units associated with the channel measurement. Alternatively, the CSI report can be associated with a second set of FD units (different from the plurality of FD units associated with the channel measurement) or/and a second set of TD units (different from the plurality of TD units associated with the channel measurement). In this later case, the UE, based on the channel measurement, can perform prediction (interpolation or extrapolation) in the second set of FD units or/and the second set of TD units associated with the CSI report.

[0198] An illustration of the SD units (in 1.sup.st and 2.sup.nd antenna dimensions), FD units, and, and TD units is shown in FIG. **14**. [0199] The first dimension is associated with the 1st antenna port dimension and comprises N.sub.1 units, [0200] The second dimension is associated with the 2nd antenna port dimension and comprises N.sub.2 units, [0201] The third dimension is associated with the frequency dimension and comprises N.sub.3 units, and [0202] The fourth dimension is associated with the time/Doppler dimension and comprises N.sub.4 units.

[0203] Alternatively, the SD units, FD units, and, and TD units are as follows. [0204] The first

dimension is associated with the antenna port dimension and comprises $P_{\text{sub.CSIRS}}$ units, [0205] The second dimension is associated with the frequency dimension and comprises $N_{\text{sub.3}}$ units, and [0206] The third dimension is associated with the time/Doppler dimension and comprises $N_{\text{sub.4}}$ units.

[0207] The plurality of SD units can be associated with antenna ports (e.g. co-located at one site or distributed across multiple sites) comprising one or multiple antenna/port groups (i.e., $N_{\text{sub.g}} \geq 1$), and dimensionalizes the spatial-domain profile of the channel measurement.

[0208] When $K=1$, there is one CMR comprising $P_{\text{sub.CSIRS}}$ CSI-RS antenna ports. [0209] When $N_{\text{sub.g}}=1$, there is one PG comprising $P_{\text{sub.CSIRS}}$ ports, and the CSI report is based on the channel measurement from the one PG. [0210] When $N_{\text{sub.g}} > 1$, there are multiple PGs, and the CSI report is based on the channel measurement from/across the multiple PGs.

[0211] When $K > 1$, there are multiple CMRs, and the CSI report is based on the channel measurement across the multiple CMRs. In one example, a CMR corresponds to an PG (one-to-one mapping). In one example, multiple CMRs can correspond to an PG (many-to-one mapping).

[0212] In one example, when the $P_{\text{sub.CSIRS}}$ antenna ports are co-located at one site, $N_{\text{sub.g}}=1$. In one example, when the $P_{\text{sub.CSIRS}}$ antenna ports are distributed (non-co-located) across multiple sites, $N_{\text{sub.g}} > 1$.

[0213] In one example, when the $P_{\text{sub.CSIRS}}$ antenna ports are co-located at one site and within a single antenna panel, $N_{\text{sub.g}}=1$. In one example, when the $P_{\text{sub.CSIRS}}$ antenna ports are distributed across multiple antenna panels (can be co-located or non-co-located), $N_{\text{sub.g}} > 1$.

[0214] The value of $N_{\text{sub.g}}$ can be configured, e.g. via higher layer RRC parameter. Or, it can be indicated via a MAC CE. Or, it can be provided via a DCI field.

[0215] Likewise, the value of K can be configured, e.g. via higher layer RRC parameter. Or, it can be indicated via a MAC CE. Or, it can be provided via a DCI field.

[0216] In one example, $K=N_{\text{sub.g}}=X$. The value of X can be configured, e.g. via higher layer RRC parameter. Or, it can be indicated via a MAC CE. Or, it can be provided via a DCI field.

[0217] In one example, the value of K is determined based on the value of $N_{\text{sub.g}}$. In one example, the value of $N_{\text{sub.g}}$ is determined based on the value of K .

[0218] The plurality of FD units can be associated with a frequency domain allocation of resources (e.g. one or multiple CSI reporting bands, each comprising multiple PRBs) and dimensionalizes the frequency (or delay)-domain profile of the channel measurement.

[0219] The plurality of TD units can be associated with a time domain allocation of resources (e.g. one or multiple CSI reporting windows, each comprising multiple time slots) and dimensionalizes the time (or Doppler)-domain profile of the channel measurement.

[0220] In one example, the number of antenna ports across K CSI-RSs (resources or CMRs) is the same. For example, each of the K CSI-RSs (resources or CMRs) can be associated with $2N_{\text{sub.1}}N_{\text{sub.2}}$ antenna ports. In this case, the total number of antenna ports is $P_{\text{sub.CSIRS,tot}}=2KN_{\text{sub.1}}N_{\text{sub.2}}$.

[0221] In one example, the number of antenna ports across K CSI-RSs (resources or CMRs) can be the same or different. For example, each of the K CSI-RSs (resources or CMRs) can be associated with $2N_{\text{sub.1},r}N_{\text{sub.2},r}$ antenna ports. In this case, the total number of antenna ports is $P_{\text{sub.CSIRS,tot}}=\sum_{r=1}^{\text{sup.}} K2N_{\text{sub.1},r}N_{\text{sub.2},r}$.

[0222] In port numbering scheme 1, the CSI-RS ports are numbered according to the order of (polarization p , NZP CSI-RS resource r) as CSI-RS ports of $(p=0, r=1)$ followed by CSI-RS ports of $(p=1, r=1)$, followed by CSI-RS ports of $(p=0, r=2)$, followed by CSI-RS ports of $(p=1, r=2)$, . . . , followed by CSI-RS ports of $(p=0, r=N)$ followed by CSI-RS ports of $(p=1, r=N)$.

[0223] In port numbering scheme 2, the CSI-RS ports are numbered according to the order of (polarization p , NZP CSI-RS resource r) as [0224] CSI-RS ports of $(p=0, r=1)$ followed by CSI-RS ports of $(p=0, r=1)$, . . . , followed by CSI-RS ports of $(p=0, r=N)$, and [0225] then CSI-RS ports of $(p=1, r=1)$ followed by CSI-RS ports of $(p=1, r=1)$, . . . , followed by CSI-RS ports of $(p=1, r=N)$.

[0226] In one example, an PG corresponds to an antenna, an antenna group (multiple antennae), an antenna port, an antenna port group (multiple ports), a CSI-RS resource, a CSI-RS resource set, a group of CSI-RS resources, a panel, an RRH, a Tx-Rx entity, a (analog) beam, a (analog) beam group, a cell, a cell group.

[0227] In one example, PGs can have a uniform (the same/common) structure. For example, they can have the same number of ports ($P_{\text{sub.CSIRS},r}=P_{\text{sub.CSIRS}}$) or the same antenna port layout ($N_{\text{sub.1},r}, N_{\text{sub.2},r}=(N_{\text{sub.1}}, N_{\text{sub.2}})$). In one example, PGs can have non-uniform (or different) structure. For example, they can have the same or different number of ports ($P_{\text{sub.CSIRS},r,\text{sub.1}}=P_{\text{sub.CSIRS},r,\text{sub.2}}$ or $P_{\text{sub.CSIRS},r,\text{sub.1}}\neq P_{\text{sub.CSIRS},r,\text{sub.2}}$) or the same antenna port layout, i.e., ($N_{\text{sub.1},r,\text{sub.1}}, N_{\text{sub.2},r,\text{sub.1}}=(N_{\text{sub.1},r,\text{sub.2}}, N_{\text{sub.2},r,\text{sub.2}})$) or ($N_{\text{sub.1},r,\text{sub.1}}, N_{\text{sub.2},r,\text{sub.1}}\neq(N_{\text{sub.1},r,\text{sub.2}}, N_{\text{sub.2},r,\text{sub.2}})$).

[0228] FIG. 15 illustrates a diagram of example PG hypotheses 1500 according to embodiments of the present disclosure. For example, PG hypotheses 1500 can be utilized by the BS 102 of FIG. 2. This example is for illustration only and other embodiments can be used without departing from the scope of the present disclosure.

[0229] In one embodiment, a user is configured with a dynamic (flex-) MIMO framework based on a single basic NW entity, namely a port group (PG) is provided. The basic NW entity can be an O-RAN RU. A PG can be an abstraction for antenna panel ($N_{\text{sub.g}}=1$), TRP ($N_{\text{sub.g}}\geq 1$), CSI-RS antenna ports of a NZP CSI-RS resource ($N_{\text{sub.g}}=1$), a subset of CSI-RS antenna ports of a NZP CSI-RS resource ($N_{\text{sub.g}}\geq 1$), CSI-RS antenna ports of multiple NZP CSI-RS resources comprising a CSI-RS resource set. The MIMO operations in the provided framework includes at least three steps: [0230] 1) NW configuring a UE (e.g., the UE 116) to measure $N_{\text{sub.g}}\geq 1$ PGs [0231] 2) The selection of at least one PG selection hypothesis [0232] 3) CSI reporting and DL-precoding according to the at least one of the PG selection hypotheses.

[0233] Let $Y\geq 1$ be a number of PG selection hypotheses. An example is shown in FIG. 15, wherein $N_{\text{sub.g}}=4$, and six examples of PG selection hypotheses are shown. For each $l\in\{1, \dots, Y\}$, an PG selection hypothesis selects $n_{\text{sub.l}}$ PGs, where $1\leq n_{\text{sub.l}}\leq N_{\text{sub.AG}}$. The PG selection can either be user-reported (performed by the user, and reported to the NW), or NW-controlled (performed by the NW, and indicated to the user), and can either be a standalone/separate or a non-standalone procedure. When non-standalone, the PG selection can be included or/and multiplexed either with a DL indication such as beam or TCI state indication, or with a report (e.g. CSI or beam report). When user-reported, the PG selection report includes an information about the selected $\{n_{\text{sub.l}}\}$ PGs, and can also include a metric (e.g. RSRP, power level, SINR). The PG selection is signaled dynamically, e.g., via a MAC CE or a DCI or a DCI with MAC CE activation (similar to beam/TCI state indication) or via UCI (similar to aperiodic beam/CSI report). Note that the PG selection can support both NW-controlled features such as beam (TCI state) indication, TRP indication (CJT), port indication (non-PMI feedback), sub-configurations (NES), and SBFD (duplexing); and User-reported features such as dynamic TRP selection (CJT), dynamic port selection (PS T2 codebook).

[0234] Depending on the antenna architecture (fully-digital vs hybrid), a PG can comprise one or more than one antenna port, wherein the port can be with a fixed beam/virtualization, or a dynamic beam/virtualization. Likewise, in mobility scenarios, the PG selection can be via layer 1 control in order to facilitate fast PG selection switch/update.

[0235] FIG. 16 illustrates a diagram of an example TCI state configuration 1600 according to embodiments of the present disclosure. For example, TCI state configuration 1600 can be configured by the BS 102 of FIG. 1. This example is for illustration only and other embodiments can be used without departing from the scope of the present disclosure.

[0236] FIG. 17 illustrates a diagram of an example TCI state configuration 1700 according to embodiments of the present disclosure. For example, TCI state configuration 1700 can be configured by the BS 103 of FIG. 1. This example is for illustration only and other embodiments

can be used without departing from the scope of the present disclosure.

[0237] In one embodiment, properties/assumptions associated with port(s) of the one or multiple PGs need to be established. Two main properties include: (a) QCL relation and (b) coherency. As for (a), it refers to channel properties that are common across ports associated with PGs, and (b) refers to transmission/reception hypothesis using multiple ports within/across PG(s). In NR MIMO, (a) and (b) are treated separately. In particular, for DL, (a) is via TCI state, and (b) is via expected transmission hypothesis (such as single-panel, multi-panel, sTRP, mTRP dynamic point selection (DPS), mTRP NCJT, mTRP CJT etc.). Likewise, for UL, (a) is via SRS resource indicator (SRI) or joint TCI state or UL TCI state, and (b) is via transmit precoding matrix indicator (TPMI). Also, there are two separate indications for Rel. 17 uTCI, i.e. UL TCI state indication via DL-DCI and TPMI indication via UL-DCI. Such a decoupled design clearly is unnecessary and inefficient, especially for layer 1 control signalling. It can be coupled into one by consolidating variables (and associated properties) into one indication framework. One such framework, namely multivariate TCI state, is provided as follows.

[0238] In one embodiment, the ports within a PG can be associated with properties regarding QCL (or TCI state) and coherency. A multivariate TCI framework for acquiring/indicating these properties is used, where the multivariate TCI state definition includes at least one or more of the following: [0239] PG selection hypotheses (e.g. for schemes DPS, NCJT, CJT, D-MIMO or seamless mobility) [0240] QCL relation [0241] Coherency within/across PGs

[0242] For QCL, relevant channel properties include (i) angular profile such as spatial filter parameter (analog beam), (ii) Delay profile such as average delay, delay spread, and (iii) Doppler profile such as Doppler shift, Doppler spread.

[0243] Quasi co-location (QCL) assumptions correspond to LT channel properties that are common across antenna ports associated with PGs. A few examples of QCL relations are as follows: [0244] {Doppler shift, Doppler spread, average delay, delay spread} [0245] {Doppler shift, Doppler spread} [0246] {Doppler shift, average delay} [0247] {Spatial filter parameter}

[0248] The following are viable options for coherency. [0249] Option 1: coherency based on new QCL-info or QCL-type (e.g. Type E) into the QCL or TCI-state definition [0250] Option 2: coherency and the PG selection hypothesis are included in the QCL or TCI-state definition [0251] Option 3: coherency is separate from QCL-info or TCI state definition. It can be a dedicated IE or RRC parameter, e.g. coherency-Info.

[0252] Examples for Option 1 and Option 2 are shown in FIG. 17 and FIG. 16, respectively, where QCL-Type A, B, C, D can be the same as defined in 5G specification (38.214) [REF 8], or they can be different, and Type E corresponds to coherency.

[0253] A measurement and reporting to enable multivariate TCI state indication can be supported. Such reporting can be aperiodic: either NW-controlled (via DCI either UL-DCI or DL-DCI or a dedicated DCI, including two-stage DCI) or UE-initiated (e.g. UCI either PUCCH or PUSCH or dedicated layer 1 UL channel, including two-stage UCI). The UE-initiated can be based on an event detection.

[0254] As mentioned below, the inter-PG mobility can be used to update a candidate set of PGs in order to maintain seamless “cell-free” access to RRC-connected UEs.

[0255] Akin to NR UL codebook, the Coherency between/within PGs can be (1) full-coherence (FC), i.e., a layer (stream) is transmitted using antenna ports, (2) partial-coherence (PC), i.e., a layer (stream) is transmitted using at least two but not each of antenna ports, or (3) non-coherence (NC), i.e., a layer (stream) is transmitted using one antenna port.

[0256] FIG. 18 illustrates a diagram of an example analog beam-based network 1800 according to embodiments of the present disclosure. For example, analog beam-based network 1800 can be implemented within the wireless network 100 of FIG. 1. This example is for illustration only and other embodiments can be used without departing from the scope of the present disclosure.

[0257] FIG. 19 illustrates examples of a UE moving on a trajectory 1900 located in co-located and

distributed PGs according to embodiments of the present disclosure. For example, UE moving on a trajectory **1900** can be implemented by any of the UEs **111-116** of FIG. **1**, such as the UE **112**. This example is for illustration only and other embodiments can be used without departing from the scope of the present disclosure.

[0258] In one embodiment, a cellular region can be served by partitioning (or covering) the region into (with) multiple sites and deploying multiple PGs at each site. An example is illustrated in FIG. **18** wherein there are three PGs per site. For a frequency f_1 (e.g. FR1), each PG is controlled by a fully-digital processing chain, implying there is no analog beam or a fixed beam, and PGs at one site together can serve users belonging to the respective site. At a higher frequency f_2 (e.g. FR2), each PG is associated with a hybrid analog-digital structure, implying each antenna port of the PG needs to be assigned/updated with one of the three wide beams. At an even higher frequency, since analog beams get narrower (reduced beam-width), number of analog beams increases, hence antenna ports of an PG need to be assigned/updated with one of the nine narrow beams. In general, a PG can be assigned/updated with $A_{\text{sub.g}}$ analog beams. When $A_{\text{sub.g}}=1$, there is one analog beam per PG. When $A_{\text{sub.g}}=N_{\text{sub.panel}}$, there is one analog beam per antenna panel/port per PG. In a mobility scenario, multiple PGs can serve a moving user. An illustration is shown in FIG. **19**. While the user moves from a location A to another location B, the set of PGs is updated from {PG2, PG3, PG4} to {PG1, PG2}. Consequently, a seamless beam-based (as opposed to cell-based) mobility is feasible especially for RRC-connected UEs.

[0259] An example of parameters relevant for PGs are tabulated in Table 0.5. Depending on carrier frequency, BF, and NW topology, the user can be configured with $N_{\text{sub.g}}$ PGs and values of relevant parameters.

TABLE-US-00004 TABLE 0 Values Parameters Fully-digital Hybrid BF Number of PGs, $N_{\text{sub.g}}$ {1, 2, 3, 4} $N_{\text{sub.panel}} \in \{1, 2, 4\}$ Number of ports per PG, $P_{\text{sub.g}}$ {1, 2, 4, . . . , 256} 1 Number of beams per port, $N_{\text{sub.b}}$ $1 \leq n_p \leq 256$ Port assignment/update Semi-static (fixed) Dynamic FD granularity T-F patterns, repeats WB TD granularity across RBs Multiple symbols (one per beam) Density {0.5, 1} {1, 3} Measurement One-shot: AP Beam-sweeping Multi-shot: P/SP (symbol-level)

[0260] In one example, for antenna ports within an PG, LT channel properties (QCL and coherency) remain the same, and for antenna ports across PGs, LT channel properties (QCL and coherency) can be different. In one example, for antenna ports within an PG or across PGs, LT channel properties (QCL and coherency) can be same or different.

[0261] The DCI payload size for indicating a multivariate TCI state can be large (due to multiple variables associated with properties). If a single DCI is used for indicating this multivariate TCI state, this will result in a large number of DCI decoding hypotheses, an issue for UE power consumption. Besides, the UE would need to buffer the control channel carrying DCI bit before decoding, adding further to the UE power consumption. It is therefore essential to design more efficient layer 1 control signalling. One such design can be based on a dual-/two-stage DCI.

[0262] In one embodiment, a dual-stage control (DCI) can also be used/configured/indicated for the NW-controlled PG selection, and the indication of the multivariate TCI state, especially when the number of activated TCI states is large (e.g. multi-beam MIMO, inter-cell multi-beam operations, coherent transmission etc.), including a fixed first stage payload size that indicates the information size and content of the second stage. For instance, for the NW-controlled PG selection, [0263] DCI part 1 includes a first information about the PG selection hypothesis, and size (payload) of the second stage DCI [0264] DCI part 2 includes a second information based on the first information Two examples can be as follows. [0265] Ex1: (1st information, 2nd information)=(number of PG selection hypotheses, indices of selected PGs) [0266] Ex2: (1st information, 2nd information)=(a set of PGs, properties of the selected PGs)

[0267] In one embodiment, a UE can be configured with the PG-based framework to support advanced features such as NW energy saving (NES), dynamic port assignment (ON/OFF) for FR2,

and MIMO for Duplex. In particular, these features can be based on the multivariate TCI state definition, e.g. via different PG selection hypotheses.

[0268] In one embodiment, in mobility scenarios, to provide seamless (un-interrupted) service to a RRC-connected UE, the MIMO framework herein can include a dynamic (layer 1)-based update of PG(s) for the UE, this update can be based measurement/reporting of PG(s) from a set of configured/activated candidate PG(s), akin to the beam measurement/reporting and TCI state indication in NR.

[0269] In one embodiment, a UE is configured (e.g. via higher layer) with a measurement procedure that includes information. [0270] Component (C1): a number of ports $N_{\text{sub.port}}$ [0271] Component (C2): a set of T-F resources (REs) S , e.g. S is a pattern (from multiple patterns) [0272] Component (C3): a mapping between C1 and C2 including [0273] (C3-1): density d =number of REs per PRB per port, e.g. 0.5, 1, 3; when $d=0.5$, a port is measured via an RE in even-numbered or odd-numbered PRBs, when $d=1$, a port is measured via an RE in each PRB; when $d=3$, a port is measured via 3 REs in each PRB, and so on. [0274] (C3-2): when $N_{\text{sub.port}} > 1$, each port $p_{\text{sub}.i}$ is assigned a subset of REs $S_{\text{sub}.i}$, where $i \in \{0, \dots, N_{\text{sub.port}}-1\}$ or $\in \{1, \dots, N_{\text{sub.port}}\}$, where the value $p_{\text{sub}.i} = X+i$ and X =port number of the first of the $N_{\text{sub.port}}$ ports.

[0275] In one example, a mapping/association of ports with (FD, TD) resources (e.g. T-F patterns or set of REs) is configured.

[0276] In one embodiment, a ULE is configured (e.g. via higher layer) with a measurement region/entity which is associated with $Z \geq 1$ RSs. For example, each RS can be NZP CSI-RS, ZP CSI-RS, CSI-IM, SSB, a new DL RS, DL DMRS, path-loss-RS (PL-RS), positioning RS (PRS), or phase tracking reference signal (PTRS) or TRS. The slot or slot offset (w.r.t. to a reference slot such as the slot in which corresponding DCI is received/decoded) for measuring the Z RSs can be fixed, or configured (RRC), or indicated via MAC CE or DCI.

[0277] In one example, the measurement region/entity is within a time slot (e.g. DL slot). [0278] In one example, the RS can only be (A1) aperiodic (AP), i.e. triggered via DCI field (e.g. CSI request field in a DCI). [0279] In one example, the RS can be (A2) AP with repetition ON/OFF. When repetition is ON, the RS can be measured over a number of repetitions in time domain (e.g. consecutive slots, or slots separated by m slots). The information about the repetition (e.g. value of m) can be configured via higher layer or indicated via the MAC CE or/and DCI. [0280] In one example, the RS can be (A3) AP and multi-shot, wherein the RS can be measured with a periodicity (p) starting from a slot depending on the slot offset. The value of p can be fixed, configured (via RRC) or indicated via DCI. [0281] In one example, the RS can be measured (A4) via a slot associated with a semi-persistent (SP) measurement entity with respect to a slot depending on the slot offset. The slot can be a first slot or a last slot or a configured slot from the set of slots associated with the SP measurement entity. [0282] In one example, the RS can be AP or SP or P (the 1.sup.st slot with SP resource or CMR, or the last slot with SP resource or CMR or the slot is configured, indicated via DCI, e.g. based in an offset). [0283] In one example, the RS can either be A1 or A2. [0284] In one example, the RS can either be A1 or A3. [0285] In one example, the RS can either be A1 or A4. [0286] In one example, the RS can either be A1 or A2 or A3. [0287] In one example, the RS can either be A1 or A2 or A4. [0288] In one example, the RS can either be A1 or A3 or A4. [0289] In one example, the RS can either be A2 or A3 or A4. [0290] In one example, the RS can either be A1 or A2 or A3 or A4.

[0291] In one example, the measurement region/entity can be within a time interval comprising K slots separated by m slots. The value of m can be fixed, e.g. $m=1$ (consecutive slots). The information about or the value of m can be configured via higher layer or indicated via the MAC CE or/and DCI. In one example, the Z RSs corresponds to a burst or window of measurement slots.

[0292] In one example, the measurement region/entity can be within a set of time slots, where the set of slots can be configured via higher layer or indicated via the MAC CE or/and DCI.

[0293] In one example, the measurement region/entity can be configured for one or multiple of the

following purpose or use cases. [0294] In one example, the measurement region/entity can only be configured for channel measurement (CMR). [0295] In one example, the measurement region/entity can only be configured for interference measurement (IMR). [0296] In one example, the measurement region/entity can be CMR or IMR, but not both based on a configuration (RRC) or a dynamic signaling (MAC CE or DCI). [0297] In one example, the measurement region/entity can be associated with a set of N.sub.port,CMR ports (or PG(s)) configured/indicated as CMR or/and a set of N.sub.port,IMR ports (or PG(s)) configured/indicated as IMR, where N.sub.port,CMR+N.sub.port,IMR=N.sub.port. In one example, the value of N.sub.port,CMR or/and N.sub.port,IMR can be fixed, or configured (e.g. RRC) or indicated (e.g. MACE CE or/and DCI). In one example, the value of N.sub.port or/and N.sub.port,CMR can be fixed, or configured (e.g. RRC) or indicated (e.g. MACE CE or/and DCI). In one example, the value of N.sub.port or/and N.sub.port,IMR can be fixed, or configured (e.g. RRC) or indicated (e.g. MACE CE or/and DCI). [0298] In one example, N.sub.port,IMR=1. [0299] In one example, a bitmap (b.sub.0 . . . b.sub.N.sub.port.sub.-1 or b.sub.N.sub.port.sub.-1 . . . b.sub.0 or b.sub.1 . . . b.sub.N.sub.port or b.sub.N.sub.port . . . b.sub.1) of length N.sub.port bits is used to indicate the assignment of ports as CMR or IMR. For instance, when a bit value=0, the corresponding port is a CMR port, and when a bit value=1, the corresponding port is a IMR port. Or, when a bit value=1, the corresponding port is a CMR port, and when a bit value=0, the corresponding port is a IMR port. [0300] In one example, the first N.sub.port,CMR ports are CMR, and the remaining N.sub.port,IMR ports are IMR. [0301] In one example, a 1-bit signaling or a parameter or a code point is used to indicate one of the two options. [0302] In one example, the two options are CMR and IMR. [0303] In one example, the two options are according to one or more examples described herein. [0304] In one example, a Y-bit signaling or a parameter or a code point is used to indicate one of the four examples herein. [0305] In one example, at least one of the following examples is used/configured for the value of N.sub.port. [0306] In one example, the UE can be dynamically indicated with a value of N.sub.port via MAC CE or DCI. [0307] In one example, an initial value of N.sub.port can be fixed or configured, but this value can be changed dynamically via MAC CE or DCI. [0308] In one example, multiple candidate values for N.sub.port are configured (RRC), and one of them is indicated via MAC CE or/and DCI as the value of N.sub.port for the measurement. [0309] In one example, at least one of the following examples is used/configured for a value of power or power offset associated with the RS or port or PG(s). [0310] In one example, the UE can be dynamically indicated with a value power offset via MAC CE or DCI. In one example, this value is the same/common for N.sub.port ports. In one example, this value is indicated for each of the N.sub.port ports. [0311] In one example, an initial value of power offset can be fixed or configured, but this value can be changed dynamically via MAC CE or DCI. [0312] In one example, multiple candidate values for power offset are configured (RRC), and one of them is indicated via MAC CE or/and DCI as the value of power offset for the measurement [0313] In one example, at least one of the following examples is used/configured regarding a number of CSI-RS ports (value of N.sub.CSI-RS) associated with a CSI reporting (e.g. that is triggered via a DCI field indication the corresponding CSI-Trigger-State that can be configured via higher layer). [0314] In one example, the number of CSI-RS ports can be changed based on the number of PGs (the value of N.sub.g) dynamically (e.g. via MAC CE or/and DCI). For example, N.sub.CSI-RS=N.sub.gN.sub.port. Or,

$$[00003] N_{\text{CSI-RS}} = \frac{N_{\text{port}}}{N_g}.$$

Here N.SUB.g.≥1.

[0315] In one example, the number of CSI-RS ports can be changed based on the number of ports associated with at least one PG r (the value of P.sub.CSIRS.r). For example, N.sub.CSI-RS,r=N.sub.gN.sub.port. Or,

$$[00004] N_{\text{CSI-RS},r} = \frac{N_{\text{port}}}{N_g}.$$

[0316] In one example, when N.sub.g>1 PGs, [0317] In one example,

the number of CSI-RS ports for CSI reporting can be indicated for each PG separately. There are $N_{\text{sub.g}}$ or $N_{\text{sub.g}}-1$ (when the last value of $N_{\text{sub.port}}$ minus sum of indicated values) values indicated. [0318] In one example, the number of CSI-RS ports for CSI reporting can be indicated across PGs. In one example, $N_{\text{sub.port}}$ =indicated value. In one example, the indicated value is distributed across PGs based on a fixed mapping or configured mapping.

[0319] One use case of this feature can be enabling faster measurement of different number of ports, or network energy saving or switching from a set of PGs to another set of PGs (seamless/RRC-connected mobility).

[0320] In one example, for a given measurement region/entity (as described herein), configured via RRC, the number of SD measurement unit (e.g. CSI-RS ports) can be assigned dynamically via DCI or/and MAC CE, for example, from 16 ports to 32 ports or vice versa. [0321] In one example, a higher layer or/and MAC CE can indicate multiple candidates for the number of SD units (e.g. CSI-RS ports), and a DCI can indicate one of the multiple candidate values. [0322] In one example, a higher layer can indicate multiple candidates for the number of SD units (e.g. CSI-RS ports), and a MAC CE or/and a DCI can indicate one of the multiple candidate values. [0323] In one example, the value of $N_{\text{sub.port}}$ and the mapping of port indices to the measurement region (i.e. C1 and C3 in one or more embodiments described herein) can be indicated dynamically via MAC CE or/and DCI. [0324] In one example, there are multiple candidate patterns for each candidate value of $N_{\text{sub.port}}$, and the UE (e.g., the UE 116) is configured with one pattern (from the candidates) and one value of $N_{\text{sub.port}}$ (from the candidates) dynamically. [0325] In one example, the indication is completely dynamic. For example, any port can be assigned any candidate RE either within a PRB or across multiple PRBs. [0326] In one example, when the measurement region is uniform in time and frequency (i.e. candidate measurement REs form a uniform pattern within a PRB or across multiple PRBs), the port assignment can also be uniform (e.g. uniformly multiplexed) starting at an RE and following a route or pattern (e.g. either first in time then in frequency or vice versa).

[0327] In one example, a UE can be configured with an AP CSI trigger state (via higher layer) for the CSI reporting associated with the measurement region (as described herein), where the CSI reporting is based on multiple (e.g. two) UL (e.g. PUSCH) slots (e.g. consecutive or separated). This can be beneficial when the CSI overhead is so large that partitioning UCI carrying the CSI into ≥ 4 parts (e.g. two-part UCI, in each of the two slots) is needed. [0328] In one example, the CSI reporting corresponds to one report that is partitioned into multiple parts, and each part is reported in one of the multiple slots. [0329] In one example, the CSI reporting corresponds to multiple reports, and each report is reported in one of the multiple slots.

[0330] In one example, in such multi-shot based CSI reporting, the measurement can be performed before the first of the multiple slots, and up to the CSI reference resource slot. In one example, for P or SP measurement, the measurement can also be performed after the CSI reference resource slot but before the respective time slot for CSI reporting.

[0331] In one example, the slot or slot offset for the multi-shot CSI reporting is according to at least one of the following examples. [0332] In one example, one slot offset is indicated (e.g. via RRC or MAC CE or DCI) and, the rest of the slots are determined from (w.r.t.) the indicated slot (e.g. consecutive or uniformly spaced slots). [0333] In one example, multiple slot offsets are indicated (e.g. via RRC or MAC CE or DCI), e.g. one per report/part. [0334] In one example, multiple slot offsets are determined based on a SP report (e.g. MAC CE or DCI triggering the periodic reporting with a period, fixed or configured). [0335] In one example, a first slot offset is indicated by NW (e.g. via RRC or MAC CE or DCI) and is associated with the first report/part, and the remaining slot offsets are based on UE-initiated trigger/report for the remaining parts/reports. [0336] In one example, the value of $N_{\text{sub.port}}$ can be a number, including or excluding one or more values supported in NR (i.e. 1, 2, 4, 8, 12, 16, 24, 32). For instance, it can be an odd value (e.g. 3, 5, 6, 7, ...). [0337] In one example, the value can be 1 or an even number $\{2, 4, \dots, M\}$ where M is max value (e.g. 32, 64, 128, 256). [0338] In one example, the value can be from $\{1, 2\}$

or {1,2,3} or a number which is a multiple of 4, i.e., {4, 8, . . . , M} where M is max value (e.g. 32, 64, 128, 256). [0339] In one example, the value can be from {1, 2, 4} or {1, 2, 3, 4} or {1, 2, 4, 6}, or {1, 2, 3, 4, 6} or {1, 2, . . . , 7} or a number which is a multiple of 8, i.e., {8, 16, . . . , M} where M is max value (e.g. 32, 64, 128, 256). [0340] In one example, the value can be from {1, 2, 4} or {1, 2, 3, 4} or {1, 2, 4, 6}, or {1, 2, 3, 4, 6} or {1, 2, . . . , 15} or a number which is a multiple of 16, i.e., {16, 32, . . . , M} where M is max value (e.g. 64, 128, 256). [0341] In one example, the value can be from {1, 2, 4} or {1, 2, 3, 4} or {1, 2, 4, 6}, or {1, 2, 3, 4, 6} or {1, 2, . . . , 31} or a number which is a multiple of 32, i.e., {32, 64, . . . , M} where M is max value (e.g. 64, 128, 256). [0342] In one example, a number of ports in a PG can be fixed/configured to one, and hence N.sub.port=N.sub.g. In one example, the value of N.sub.g can be configured (RRC) or dynamically (MACE CE or/and DCI) from a set of candidate values. The candidate values can belong to {1, 2, 3, 4}, or {1, 2, . . . , 8}, or {1, 2, . . . , M} and the value M can be from {4, . . . , 64} or {4, . . . , 128} or {4, . . . , 256}.

[0343] In one example, the DCI used to indicate/trigger the measurement entity/region itself or a parameter/components associated with it (as described herein) can be according to one of the following examples. [0344] In one example, the DCI is DL-related DCI (with or without DL PDSCH assignment) or a dedicated DL DCI. [0345] In one example, the DCI is UL-related DCI (with or without UL PUSCH grant) or a dedicated UL DCI. [0346] In one example, the DCI is a multivariate DCI or DCI indicating a multivariate TCI, where multivariate implies several variable number of parameters. [0347] In one example, the DCI is two-stage DCI with stage 1 DCI having a small fixed payload size, and indicating the size of the stage 2 DCI. Both stages of the DCI are in the same slot. [0348] In one example, the DCI is two-stage multivariate DCI with stage 1 multivariate DCI having a small fixed payload size, and indicating the size of the stage 2 DCI. Both stages of the DCI are in the same slot.

[0349] In one example, a UE is indicated with an indication (code point or DCI field) including a value of N.sub.port via a DCI-common (which can be a UE-group-common DCI common for multiple UEs), and the indication assigns N.sub.port to the configured measurement entity. The UE can then be triggered with an AP CSI report that is associated with the measurement region. This trigger can be from the NW (e.g. via a DCI, same or different from the DCI-common). Or, the UE can itself trigger a UE-initiated CSI reporting (e.g. based on an event detection).

[0350] In one example, [0351] In one example, when number of ports is reduced (e.g. halved) via dynamic indication, the set of candidate REs for measurement also reduces (by the same or different fraction/amount), and vice versa. [0352] In one example, the set of candidate REs remains the same regardless of whether number of ports is changes via dynamic indication. In this case, the density of the measurement (of a port) increases/decreases depending on the indicated value of number of ports N.sub.port.

[0353] In one embodiment, a UE receives a configuration (e.g. CSI-RS-Config) including one of the following: [0354] In one example, a list of CSI Trigger States, and MAC-CE or/and DCI triggers one [0355] In one example, a list of CSI Trigger States, each may include an ID or a list of multiple sub-configurations; and MAC-CE or/and DCI triggers one [0356] In one example, 1 CSI Trigger State [0357] In one example, 1 CSI Trigger State and one or multiple sub-configurations within that CSI Trigger State [0358] MAC CE activates 1 Trigger State; and DCI indicates one or multiple sub-configs [0359] In one example, >1 CSI Trigger State and one or multiple sub-configurations within each CSI Trigger State [0360] MAC CE activates 1 Trigger State; and DCI indicates one or multiple sub-configs [0361] In one example, multiple configurations (e.g. CSI Trigger States) [0362] MAC CE or/and DCI triggers one Trigger State

[0363] In one embodiment, a UE receives a configuration (e.g. CSI-RS-Config), where (FD, TD) higher layer configured (as described herein) and SD (ports or PGs) via MAC CE or DCI (as described herein). [0364] In one example, Pattern repeat in each PRB or across multiple PRBs [0365] In one example, Multiple PRBs can on consecutive [0366] In one example, Multiple can be

comb-like uniformly separated [0367] In one example, there is no pattern, a rule is used instead [0368] The present disclosure relates generally to wireless communication systems and, more specifically, to measurement in next generation of MIMO systems.

TABLE-US-00005 TABLE 1 Definition of frequency ranges Frequency range designation

Corresponding frequency range FR1 410 MHz-7125 MHz FR2 FR2-1 24250 MHz-52600 MHz
FR2-2 52600 MHz-71000 MHz

[0369] In one example, a TRP can be functionally equivalent to (hence can be replaced with) or is interchangeable with one of more of the following: an antenna, or an antenna group (multiple antennae), an antenna port, an antenna port group (multiple ports), a CSI-RS resource, multiple CSI-RS resources, a CSI-RS resource set, multiple CSI-RS resource sets, an antenna panel, multiple antenna panels, a Tx-Rx entity, a (analog) beam, a (analog) beam group, a cell, a cell group.

[0370] A NW can be built upon a spatial resource entity, say X (e.g. TRP or O-RU). For FR1/FR3, the spatial entity X comprises one or more antenna ports that are fully-digital (i.e. each antenna port is driven by a dedicated baseband processing chain), as shown in FIG. 12; and in FR2, the entity X comprises one or more antenna panels (sub-arrays), each comprising one (or two) antenna ports that is (are) controlled by analog phase shifters that result in an analog beam (pointing in certain spatial direction), as shown in FIG. 5. An antenna port in FR1/FR3 can also be beam-formed (aka virtualization); however, such a beamforming (BF) is generally static (non-adaptive, hence does not requiring measurement and reporting). In FR2, due to large propagation loss at mmWave frequencies, each antenna panel requires dynamic frequent update of the analog BF, which is often based on (analog) beam measurement and reporting.

[0371] A communication between NW and a user can broadly be based on: (A1) NW resources, and (A2) signaling components, where the former corresponds to spatial-domain, frequency-domain, and time-domain (SD, FD, TD) resources allocated to the user for the communication, and the latter corresponds to components that are signaled over the NW resources. The SD resources can be based on a single TRP/O-RU (sTRP) or multiple TRPs/O-RUs (mTRP), where mTRP can be (B1) co-located at a site/location or (B2) non-co-located/distributed at multiple sites/locations, where the latter corresponds to a distributed SD resource, hence the corresponding communication hypothesis can be (C1) non-coherent joint transmission (NCJT) where a data stream (layer) is transmitted from one of the mTRPs, or (C2) coherent JT (CJT), where a data stream (layer) can be transmitted from multiple of the mTRPs. The FD resources can comprise a set of PRBs, and the TD resources can comprise one or multiple time slots (i.e., 1 slot=N.sub.sym consecutive symbols).

[0372] The signaling components include signaling associated with (D1) measurement, (D2) channel state information (CSI) report, and (D3) DL reception or UL transmission.

[0373] For (D1), the user measures channel measurement RSs (CMRs), e.g. CSI-RS, to estimate the channel condition between the sTRP/mTRP and the user. In case of sTRP, the user can measure a set comprising one or multiple DL measurement ports (or resources). For mTRP, the measurement ports (or resources) can be (E1) one set comprising one group of ports (or resources) per TRP, or (E2) one set per TRP. The user can also measure the interference based on interference measurement RSs (IMRs). A CMR can correspond to an analog beam, and can be repeated in multiple symbols for determining user's analog beam. For UL, the user transmits UL CMRs, e.g. SRS, for the NW to perform UL channel measurement and UL CSI acquisition.

[0374] For (D2), the user, based on the measurement, determines the CSI and reports it to the NW, where the CSI can be (F1) (analog) beam-related CSI, or (F2) (digital) non-beam-related CSI. For (F1), the user can determine one or multiple pairs (indicator, metric), where the indicator indicates a CMR and the metric indicates a (beam) quality (e.g. RSRP, SINR).

[0375] For (F2), a low-resolution (Type-I) CSI and a high-resolution (Type-II) CSI can be supported. The Type-I CSI can be based on L=1 DFT SD vector per layer. The Type-II CSI is based on a weighted linear combination L>1 SD DFT vectors where the weights correspond to

coefficients. The FD DFT vectors can also be introduced to reduce the CSI feedback overhead by compressing channel coefficients in both SD and FD. Type-I or Type-II CSI can also be extended to support NCJT or CJT CSI from mTRP or/and for high/medium user velocities by exploiting time-domain correlation or Doppler-domain information.

[0376] A unified transmission configuration indication (uTCI) framework supports signaling of a unified TCI state to a user, where the unified TCI state can be a DL-TCI, an UL-TCI or a joint TCI (J-TCI) state, where a DL-TCI state is applied for receiving DL channels/signals, an UL-TCI state is applied for transmitting UL channels/signals, and a J-TCI state is applied for both DL and UL channels/signals. The uTCI framework is designed to support DL receptions and UL transmissions (i) with a joint (common) beam indication for DL and UL by leveraging beam correspondence (reciprocity between DL and UL), and (ii) with separate beam indications for DL and UL, for example to mitigate maximum permissible exposure, where the beam direction of an UL transmission is different from the beam direction of a DL reception to avoid exposure of the human body to radiation.

[0377] The uTCI framework can support a beam-level mobility, known as inter-cell BM (ICBM). In ICBM, the user-dedicated channels can be configured to use a beam (i.e., TCI state) associated with a (non-serving) cell having a physical cell identity (PCI) that is different from the PCI of the serving cell. This allows fast beam-switch to a non-serving cell for user-dedicated channels at a lower layer without involving a higher layer and without incurring latency and overhead of handover.

[0378] The ICBM can be followed by a complete cell-switch triggered by lower layers, which is known as lower-layer triggered mobility (LTM). In LTM, the NW can acquire beam measurements, and UL timing information for target candidate cells before cell-switch. The lower layers of the NW decide when to perform a cell-switch, and send a medium access control channel element (MAC CE) containing a cell-switch command (CSC) that triggers the cell-switch from a source cell to a target cell. The CSC includes beam (i.e., TCI state) and UL timing information for the user to use on the target cell. After a beam application delay, the user and the NW communicates via the target cell.

[0379] Full-duplex transmission and reception in the same channel BW or using non-contiguous intra-band carrier aggregation (CA) is a promising technology to enhance UL coverage, reduce latency and improve system capacity and to overcome limitations inherent to the use of de-facto mandated semi-static TDD UL-DL frame configurations in today's TDD deployments. In NW-side subband full-duplex (SBFD) mode, simultaneous transmissions and receptions by the NW on the same time-domain symbol on the NR carrier can occur in non-overlapping UL and DL subbands. The users with support for NW-side SBFD operation still operate in half-duplex, i.e., the user can either transmit or receive on an SBFD symbol but not transmit and receive simultaneously. An SBFD UL subband can be located in the center or at the edge of the NR carrier in FR1 or FR2-1. For CA-based SBFD in FR2-1, one component carrier (CC) can be allocated for UL transmissions whereas the remaining CCs can be used for DL transmission. NW-side self-interference cancellation (SIC) capability to enable SBFD can be realized through a combination of solutions. For example, the NW can use Tx/Rx antenna isolation on the antenna panel(s), beam steering, analog and/or digital pre-distortion, digital interference cancellation, and analog and/or digital filtering solutions.

[0380] As explained, the 5G NW can support several features, services, use cases, and deployment scenarios. It however also introduces too many different abstractions (for specification) of NW entities and involved signaling for components of these abstractions. A direct scaling/extension/reuse of these typical up to 5G solutions for 6G will add to the complexity, which is undesired in real NW deployments. In next-gen MIMO systems (e.g. 6G), therefore, at least two aspects need to be evaluated. [0381] (A) A single band-agnostic NW entity, namely a port, and an associated QCL information/relation (e.g. via a TCI state) with a source RS (or analog beam). For

CSI, multiple ports or a port group (PG) (as a collection of ports) can be used. [0382] (B) NW architecture as perceived in O-RAN: The functionality split among O-RAN entities for DL and UL operations, such as O-RU, O-DU, and O-CU (as described herein). An example is shown in FIG. 11. In particular, the PHY functionality split between O-DU and O-RU includes at least the following aspects. [0383] (B1) PHY processing: [0384] bit-level processing, [0385] symbol-level processing [0386] (B2) Scheduling (residing in MAC): SU-MIMO/MU-MIMO scheduling across different O-RUs or/and allocated frequency-domain resources (e.g. PRBs, PRGs, SBs) [0387] Utilizing UCI carrying CSI [0388] If DL/UL reciprocity is feasible, also utilizing SRS-based channel measurement [0389] (B3) Precoder calculation at a gNB (NW side) for DL-SCH transmission: [0390] For SU-MIMO, precoder can simply follow the PMI (calculated assuming SU-MIMO hypothesis) reported by the UE, or, if DL/UL reciprocity is feasible, be calculated from the eigenvector(s) of the measured DL channels. [0391] For MU-MIMO, precoder needs to be calculated based on additional orthogonalization (e.g. ZFBF, SLNR) among PMIs, or, if DL/UL reciprocity is feasible, the eigenvectors of the measured channels of the co-scheduled UEs [0392] A few relevant (more-probable) candidates discussed in the O-RAN Alliance (depicted in FIG. 11) are shown in Table 2.

TABLE-US-00006 TABLE 2 (both DL and UL) Hi- Low- PDCP RLC MAC PHY PHY RF HLS LLS O-RAN.sup.1 O-CU: O-DU: RLC, MAC, O-RU: Low- Y symbol- (Opt7-2x) PDCP Hi-PHY PHY, RF level PHY Opt7-3 O-CU: O-DU: RLC, MAC, O-RU: Low- Y bit-level PDCP Hi-PHY PHY, RF PHY Opt8 DU: RLC, MAC, PHY RU: Y CPRI RF O-RAN.sup.1: [REF 12] [0393] Cat-A, Cat-B [0394] UL: Cat-C

[0395] The main difference between a FR1/FR3 port and a FR2 panel is that the beam/virtualization (i.e. port assignment) is fixed in the former, and it requires (a) measurement and reporting from UE (e.g., the UE 116) and (b) a beam indication from the NW (e.g. TCI state with QCL-TypeD source RS). It is therefore plausible to have a unified framework in which a port in FR1 and a panel in FR2 can be abstracted based on a unified, band-agnostic spatial entity, e.g. port or port group (PG), and associated QCL and coherency properties across ports or PGs (intra-/inter PG). For instance, [0396] Port: a FR1/FR3 port and a FR2 beam/source [0397] FR2: multiple ports/beams for CSI acquisition and DL/UL transmission [0398] FR1/FR3: a port group (PG), a collection/set/group of ports for CSI acquisition and DL/UL transmission [0399] While the O-RAN Alliance is intended for 5G NR, it is expected that its framework will continue, or at most refined, for 6G. The O-RAN Alliance specifies 3 levels of functional splits—namely CU, DU, and RU—to facilitate multi-vendor inter-operability within a NW. The manner in which PHY-layer functions are split between DU and RU(s) imposes serious impact on the feasibility, performance, and complexity of different MMO schemes—mainly due to the latency and quantization loss incurred by the O-RAN-standardized RU-DU interface. [0400] In this sense, a “port” can be associated with a digital port in FR1/FR3 or an analog beam in FR2 (thereby abandoning the 5G association between an analog beam and a CSI-RS resource for FR2).

[0401] FIG. 20 illustrates a flowchart of an example BS procedure 2000 for precoding according to embodiments of the present disclosure. For example, procedure 2000 can be performed by the BS 102 in FIG. 2. This example is for illustration only and other embodiments can be used without departing from the scope of the present disclosure.

[0402] The procedure begins in 2010, a BS determines whether DL/UL reciprocity is feasible. If DL/UL reciprocity is not feasible, then in 2020, a PMI-based scheme may be performed. If DL/UL reciprocity is feasible, then in 2030, a SRS-based precoding may be performed. In 2040, the BS determines whether SRS fails. If SRS does not fail, the procedure returns to 2030. If SRS does fail, the procedure returns to 2020.

[0403] The precoding in a DL transmission scheme can be SRS-based or PMI-based, as shown in FIG. 20, where the SRS-based scheme can only use used when the DL-UL reciprocity is feasible

and SRS channel measurement is feasible (i.e. UL signal-to-noise ratio (SNR) of channel measurement is not too small). The PMI-based scheme can be used regardless of reciprocity assumptions. Both SRS-based and PMI-based precoding is determined based on a measurement (e.g. SRS-based channel measurement by NW, and CSI-RS/CSI-IM based channel/interference measurement by UE).

[0404] For a given (k, l) the set of code-division multiplexing (CDM) REs: $\{(k+k', l+l')\}$ for $\forall k', l'$, and the CDM group index: j.

TABLE-US-00007 TABLE 7.4.1.5.3-1 CSI-RS locations within a slot. Ports Density cdm- CDM group Row X ρ Type (k, l) index j k' l' #REs CDM groups 1 1 3 noCDM (k.sub.0, l.sub.0), (k.sub.0 + 4, 0, 0, 0 0 0 3 l.sub.0), (k.sub.0 + 8, l.sub.0) 2 1 1, 0.5 noCDM (k.sub.0, l.sub.0), 0 0 0 1 3 2 1, 0.5 fd- (k.sub.0, l.sub.0), 0 0, 1 0 2 (k.sub.0, l.sub.0), CDM2 (k.sub.0 + 1, l.sub.0) 4 4 1 fd- (k.sub.0, l.sub.0), (k.sub.0 + 0, 1 0, 1 0 4 $\{(k.sub.0, l.sub.0), CDM2 2, l.sub.0)\}$ (k.sub.0 + 1, l.sub.0) $\{(k.sub.0 + 2, l.sub.0), (k.sub.0 + 3, l.sub.0)\}$ 5 4 1 fd- (k.sub.0, l.sub.0), 0, 1 0, 1 0 4 $\{(k.sub.0, l.sub.0), CDM2 (k.sub.0, l.sub.0 + 1) (k.sub.0 + 1, l.sub.0)\}$ $\{(k.sub.0, l.sub.0 + 1), (k.sub.0 + 1, l.sub.0 + 1)\}$ 6 8 1 fd- (k.sub.0, l.sub.0), (k.sub.1, l.sub.0), 0, 1, 2, 3 0, 1 0 8 CDM2 (k.sub.2, l.sub.0), (k.sub.3, l.sub.0) 7 8 1 fd- (k.sub.0, l.sub.0), 0, 1, 2, 3 0, 1 0 8 CDM2 (k.sub.1, l.sub.0), (k.sub.0, l.sub.0 + 1), (k.sub.1, l.sub.0 + 1) 8 8 1 cdm4- (k.sub.0, l.sub.0), (k.sub.1, l.sub.0) 0, 1 0, 1 0, 1 8 FD2- TD2 9 12 1 fd- (k.sub.0, l.sub.0), (k.sub.1, l.sub.0), 0, 1, 2, 3, 4, 5 0, 1 0 12 CDM2 (k.sub.2, l.sub.0), (k.sub.3, l.sub.0), (k.sub.4, l.sub.0), (k.sub.5, l.sub.0) 10 12 1 cdm4- (k.sub.0, l.sub.0), (k.sub.1, l.sub.0), 0, 1, 2 0, 1 0, 1 12 FD2- (k.sub.2, l.sub.0) TD2 11 16 1, 0.5 fd- (k.sub.0, l.sub.0), (k.sub.1, l.sub.0), 0, 1, 2, 3, 0, 1 0 16 CDM2 (k.sub.2, l.sub.0), (k.sub.3, l.sub.0), 4, 5, 6, 7 (k.sub.0, l.sub.0 + 1), (k.sub.1, l.sub.0 + 1), (k.sub.2, l.sub.0 + 1), (k.sub.3, l.sub.0 + 1) 12 16 1, 0.5 cdm4- (k.sub.0, l.sub.0), (k.sub.1, l.sub.0), 0, 1, 2, 3 0, 1 0, 1 16 FD2- (k.sub.2, l.sub.0), (k.sub.3, l.sub.0) TD2 13 24 1, 0.5 fd- (k.sub.0, l.sub.0), (k.sub.1, l.sub.0), 0, 1, 2, 3, 4, 5, 0, 1 0 24 CDM2 (k.sub.2, l.sub.0), 6, 7, 8, 9, 10, 11 (k.sub.0, l.sub.0 + 1), (k.sub.1, l.sub.0 + 1), (k.sub.2, l.sub.0 + 1), (k.sub.0, l.sub.1), (k.sub.1, l.sub.1), (k.sub.2, l.sub.1), (k.sub.0, l.sub.1 + 1), (k.sub.1, l.sub.1 + 1), (k.sub.2, l.sub.1 + 1) 14 24 1, 0.5 cdm4- (k.sub.0, l.sub.0), (k.sub.1, l.sub.0), 0, 1, 2, 3, 4, 5 0, 1 0, 1 24 FD2- (k.sub.2, l.sub.0), (k.sub.0, l.sub.1), TD2 (k.sub.1, l.sub.1), (k.sub.2, l.sub.1) 15 24 1, 0.5 cdm8- (k.sub.0, l.sub.0), (k.sub.1, l.sub.0), 0, 1, 2 0, 1 0, 1, 24 FD2- (k.sub.2, l.sub.0) 2, 3 TD4 16 32 1, 0.5 fd- (k.sub.0, l.sub.0), (k.sub.1, l.sub.0), 0, 1, 2, 3, 4, 5, 6, 0, 1 0, 1 32 CDM2 (k.sub.2, l.sub.0), (k.sub.3, l.sub.0), 7, 8, 9, 10, 11, (k.sub.0, l.sub.0 + 1), 12, 13, 14, 15 (k.sub.1, l.sub.0 + 1), (k.sub.2, l.sub.0 + 1), (k.sub.3, l.sub.0 + 1), (k.sub.0, l.sub.1), (k.sub.1, l.sub.1), (k.sub.2, l.sub.1), (k.sub.3, l.sub.1), (k.sub.0, l.sub.1 + 1), (k.sub.1, l.sub.1 + 1), (k.sub.2, l.sub.1 + 1), (k.sub.3, l.sub.1 + 1) 17 32 1, 0.5 cdm4- (k.sub.0, l.sub.0), (k.sub.1, l.sub.0), 0, 1, 2, 3, 4, 5, 0, 1 0, 1 32 FD2- (k.sub.2, l.sub.0), (k.sub.3, l.sub.0), 6, 7 TD2 (k.sub.0, l.sub.1), (k.sub.1, l.sub.1), (k.sub.2, l.sub.1), (k.sub.3, l.sub.1) 18 32 1, 0.5 cdm8- (k.sub.0, l.sub.0), (k.sub.1, l.sub.0), 0, 1, 2, 3 0, 1 0, 1, 32 FD2- (k.sub.2, l.sub.0), (k.sub.3, l.sub.0) 2, 3 TD4

TABLE-US-00008 TABLE 7.4.1.5.3-2 The sequences w.sub.f (k') and w.sub.t (l') for cdm-Type equal to 'noCDM'. Index w.sub.f(0) w.sub.t(0) 0 1 1

TABLE-US-00009 TABLE 7.4.1.5.3-3 The sequences w.sub.f (k') and w.sub.t (l') for cdm-Type equal to 'fd-CDM2'. Index [w.sub.f(0) w.sub.f(1)] w.sub.t(0) 0 [+1 +1] 1 1 [+1 -1] 1

TABLE-US-00010 TABLE 7.4.1.5.3-4 The sequences w.sub.f (k) and w.sub.t(l') for cdm-Type equal to 'cdm4-FD2-TD2'. Index [w.sub.f(0) w.sub.f(1)] [w.sub.t(0) [w.sub.t(1)] 0 [+1 +1] [+1 +1] 1 [+1 -1] [+1 +1] 2 [+1 +1] [+1 -1] 3 [+1 -1] [+1 -1]

TABLE-US-00011 TABLE 7.4.1.5.3-5 The sequences w.sub.f(k') and w.sub.t(l') for cdm-Type equal to 'cdm8-FD2-TD4'. Index [w.sub.f(0) w.sub.f(1)] [w.sub.t(0) w.sub.t(1) w.sub.t(2) w.sub.t(3)] 0 [+1 +1] [+1 +1 +1 +1] 1 [+1 -1] [+1 +1 +1 +1] 2 [+1 +1] [+1 -1 +1 -1] 3 [+1 -1] [+1 -1 +1 -1] 4 [+1 +1] [+1 +1 -1 -1] 5 [+1 -1] [+1 +1 -1 -1] 6 [+1 +1] [+1 -1 -1 +1] 7 [+1 -1] [+1 -1 -1 +1]

[0405] Embodiments of the present disclosure recognizes that a CMR/IMR configuration (via higher layer signaling) has at least the following issues/drawbacks in 5G NR. [0406] T-F REs and

port mapping are coupled/configured via RRC, implying CMR/TIR update requires RRC (re-)configuration. [0407] In case of energy saving (NES), the T-F resource utilization can be inefficient (under-/non-utilized), especially when number of ports reduces (e.g. sub-configuration with portIndicatorBitmap). [0408] CSI-IM is a special configuration of ZP CSI-RS (without port-related information), hence is duplicative. [0409] Finally, the port-related info for ZP CSI-RS is needed (hence is redundant).

[0410] This disclosure provided embodiments to address these issues and also provides a measurement procedure based on decoupled/separate signaling for T-F resources and port-related information.

[0411] The present disclosure relates to next generation MIMO communication systems (e.g. adv. 5G and 6G). This disclosure, in particular provides, a measurement procedure associated with a report or a feature (not requiring a report) necessary for communication (e.g. feature tracking of time/frequency/phase); and following are aspects of this disclosure: [0412] Mapping between measurement region (set of resources) and an entity, where [0413] Entity: (a) value(s) of N.sub.port, (b) RS(s), or (c) RSs and value(s) of N.sub.port associated with each RS [0414] Density: a given RS can be associated with one of multiple density values ρ (e.g. 1, 2) depending on a value of N.sub.port; for example, (ρ , N.sub.port) can be (1, v) and

[00005](2, $\frac{v}{2}$). [0415] Example of mapping: pattern (e.g. 5G NR patterns) [0416] For DL, measurement region=ZP or NZP or ZP+NZP [0417] ZP used for PDSCH rate matching (e.g. UE does not expect PDSCH on those REs) [0418] ZP used for IMR (CSI-IM is replaced with ZP or CSI-IM in addition to ZP for IMR) [0419] NZP used for CMR [0420] NZP used for IMR (for mTRP scenarios DPS/NCJT/CJT) [0421] DCI-based switch between ZP/NZP [0422] DL-DCI: ZP trigger or one of its codepoint can be used for [0423] UL-DCI: CSI trigger [0424] Dedicated DCI [0425] Measurement for NES, CLI

[0426] Aspects, features, and advantages of the disclosure are readily apparent from the following detailed description, simply by illustrating a number of particular embodiments and implementations, including the best mode contemplated for carrying out the disclosure. The disclosure is also capable of other and different embodiments, and its several details can be modified in various obvious respects, all without departing from the spirit and scope of the disclosure. Accordingly, the drawings and description are to be regarded as illustrative in nature, and not as restrictive. The disclosure is illustrated by way of example, and not by way of limitation, in the figures of the accompanying drawings.

[0427] In the following, for brevity, both FDD and TDD are provided as the duplex method for both DL and UL signaling.

[0428] Although exemplary descriptions and embodiments to follow expect orthogonal frequency division multiplexing (OFDM) or orthogonal frequency division multiple access (OFDMA), this disclosure can be extended to other OFDM-based transmission waveforms or multiple access schemes such as filtered OFDM (F-OFDM).

[0429] This disclosure of disclosure covers several components which can be used in conjunction or in combination with one another, or can operate as standalone schemes.

[0430] All the following components and embodiments are applicable for UL transmission with CP-OFDM (cyclic prefix OFDM) waveform as well as DFT-SOFDM (DFT-spread OFDM) and SC-FDMA (single-carrier FDMA) waveforms. Furthermore, the following components and embodiments are applicable for UL transmission when the scheduling unit in time is either one subframe (which can include one or multiple slots) or one slot.

[0431] In the present disclosure, the frequency resolution (reporting granularity) and span (reporting bandwidth) of a reporting (e.g. CSI or UE assistance information) can be defined in terms of frequency “subbands” and “CSI reporting band” (CRB), respectively.

[0432] A subband for the reporting is defined as a set of contiguous PRBs which represents the smallest frequency unit for the reporting. The number of PRBs in a subband can be fixed for a

given value of DL system bandwidth, configured either semi-statically via higher-layer/RRC signaling, or dynamically via L1 DL control signaling or MAC control element (MAC CE). The number of PRBs in a subband can be included in the reporting setting or configuration.

[0433] “A reporting band” is defined as a set/collection of subbands, either contiguous or non-contiguous, wherein the reporting is performed. For example, the reporting band can include the subbands within the system bandwidth (e.g. DL BW). This can also be termed “full-band”. Alternatively, the reporting band can include only a collection of subbands within the system bandwidth. This can also be termed “partial band”.

[0434] The term “a reporting band” is used only as an example for representing a function. Other terms such as “a reporting subband set” or “a reporting bandwidth” or bandwidth part (BWP) can also be used.

[0435] In terms of UE configuration, a UE can be configured with at least one reporting band (e.g. CSI reporting band or UE assistance information reporting band). This configuration can be semi-static (via higher-layer signaling or RRC) or dynamic (via MAC CE or L1 DL control signaling). When configured with multiple (N) CSI reporting bands (e.g. via RRC signaling), a UE can report CSI associated with $n \leq N$ CSI reporting bands. For instance, >6 GHz, large system bandwidth may require multiple CSI reporting bands. The value of n can either be configured semi-statically (via higher-layer signaling or RRC) or dynamically (via MAC CE or L1 DL control signaling). Alternatively, the UE can report a recommended value of n via an UL channel.

[0436] Therefore, CSI parameter frequency granularity can be defined per CSI reporting band as follows. A CSI parameter is configured with “wideband” or “single” reporting for the CSI reporting band with M.sub.n subbands when one CSI parameter for the M.sub.n subbands within the CSI reporting band. A CSI parameter is configured with “subband” for the CSI reporting band with M.sub.n subbands when one CSI parameter is reported for each of the M.sub.n subbands within the CSI reporting band.

[0437] In the following, N.sub.1 and N.sub.2 are the number of antenna ports with the same polarization in the first and second dimensions, respectively. For 2D antenna port layouts, $N_{sub.1} > 1$, $N_{sub.2} > 1$, and for 1D antenna port layouts $N_{sub.1} > 1$ and $N_{sub.2} = 1$ or $N_{sub.2} > 1$ and $N_{sub.1} = 1$. In the rest of the disclosure, 1D antenna port layouts with $N_{sub.1} > 1$ and $N_{sub.2} = 1$ is provided. The disclosure, however, is applicable to the other 1D port layouts with $N_{sub.2} > 1$ and $N_{sub.1} = 1$. Also, in the rest of the disclosure, $N_{sub.1} \geq N_{sub.2}$. The disclosure, however, is applicable to the case when $N_{sub.1} < N_{sub.2}$, and the embodiments for $N_{sub.1} > N_{sub.2}$ apply to the case $N_{sub.1} < N_{sub.2}$ by swapping/switching ($N_{sub.1}$, $N_{sub.2}$) with ($N_{sub.2}$, $N_{sub.1}$). For a single-polarized (or co-polarized) antenna port layout, the total number of antenna ports is $P_{sub.tot} = N_{sub.1} N_{sub.2}$. And, for a dual-polarized antenna port layout, the total number of antenna ports is $P_{sub.tot} = 2 N_{sub.1} N_{sub.2}$. An illustration is shown in FIG. 13 where “X” represents two antenna polarizations (dual-pol, $s=2$) and “/” represents one antenna polarization (co-pol, $s=1$). In this disclosure, the term “polarization” refers to a group of antenna ports with the same polarization. For example, for DL, antenna ports $j = X+0, X+1, \dots$,

[00006] $X + \frac{P_{tot}}{2} - 1$

comprise a first antenna polarization, and antenna ports

[00007] $j = X + \frac{P_{tot}}{2}, X + \frac{P_{tot}}{2} + 1,$

$\dots, X + P_{sub.tot} - 1$ comprise a second antenna polarization, where $P_{sub.tot} = P_{sub.CSIRS}$ is a number of CSI-RS antenna ports and X is a starting antenna port number (e.g. $X=3000$, then antenna ports are 3000, 3001, 3002, $\dots$). Likewise, for UL, antenna ports $j = Y+0, Y+1, \dots$,

[00008] $X + \frac{P_{tot}}{2} - 1$

comprise a first antenna polarization, and antenna ports

[00009] $j = Y + \frac{P_{tot}}{2}, Y + \frac{P_{tot}}{2} + 1,$

$\dots, Y + P_{sub.tot} - 1$ comprise a second antenna polarization, where $P_{sub.tot} = P_{sub.SRS}$ is a number of SRS antenna ports and X is a starting antenna port number. Unless stated otherwise, dual-

polarized antenna layouts are expected in this disclosure. The embodiments (and examples) in this disclosure however are general and are applicable to single-polarized antenna layouts as well.

[0438] Let s denotes the number of antenna polarizations (or groups of antenna ports with the same polarization). Then, for co-polarized antenna ports, $s=1$, and for dual- or cross (X)-polarized antenna ports $s=2$. So, the total number of antenna ports $P_{\text{sub.tot}}=sN_{\text{sub.1}}N_{\text{sub.2}}$.

[0439] Let $N_{\text{sub.g}}$ be a number of ports or antenna/port groups (PGs). When there are multiple antenna/port groups ($N_{\text{sub.g}}>1$), each group ($g \in \{1, \dots, N_{\text{sub.g}}\}$) comprises $N_{\text{sub.1,g}}$ and $N_{\text{sub.2,g}}$ ports in two dimensions. This is illustrated in FIG. 13. Note that the antenna port layouts may be the same ($N_{\text{sub.1,g}}=N_{\text{sub.1}}$ and $N_{\text{sub.2,g}}=N_{\text{sub.2}}$) in different antenna/port groups, or they can be different across antenna/port groups. For group g , the number of antenna ports is $P_{\text{sub.tot,g}}=N_{\text{sub.1,g}}N_{\text{sub.2,g}}$ or $2N_{\text{sub.1,g}}N_{\text{sub.2,g}}$ (for co-polarized or dual-polarized respectively), i.e., $P_{\text{sub.tot,g}}=s_{\text{sub.g}}N_{\text{sub.1,g}}N_{\text{sub.2,g}}$ where $s_{\text{sub.g}}=1$ or 2 .

[0440] In one example, an antenna/port group corresponds to an antenna panel. In one example, an antenna/port group corresponds to a TRP. In one example, an antenna/port group corresponds to an RRH. In one example, an antenna/port group corresponds to CSI-RS antenna ports (of a PG or NZP CSI-RS resource). In one example, an antenna/port group corresponds to a subset of CSI-RS antenna ports (of a PG or NZP CSI-RS resource) (comprising multiple antenna/port groups). In one example, an antenna/port group corresponds to CSI-RS antenna ports of multiple NZP CSI-RS resources (e.g. comprising a CSI-RS resource set). In one example, an antenna/port group corresponds to SRS antenna ports (of a PG or SRS resource). In one example, an antenna/port group corresponds to a subset of SRS antenna ports (of a PG or a SRS resource) (comprising multiple antenna/port groups). In one example, an antenna/port group corresponds to SRS antenna ports of PGs or multiple SRS resources (e.g. comprising a SRS resource set).

[0441] In one example, an antenna/port group corresponds to a reconfigurable intelligent surface (RIS) in which the antenna/port group can be (re-)configured more dynamically (e.g. via MAC CE or/and DCI). For example, the number of antenna ports associated with the antenna/port group can be changed dynamically.

[0442] In one example, the antenna architecture of the MIMO system is structured. For example, the antenna structure at each PG or O-RU (or RU) is dual-polarized (single or multi-panel as shown in FIG. 13). The antenna structure at each PG or O-RU (or RU) can be the same. Or the antenna structure at an PG or O-RU (or RU) can be different from another PG or O-RU (or RU). Likewise, the number of ports at each PG or O-RU (or RU) can be the same. Or the number of ports at one PG or O-RU (or RU) can be different from another PG or O-RU (or RU).

[0443] In another example, the antenna architecture of the MIMO system is unstructured. For example, the antenna structure at one PG or O-RU (or RU) can be different from another PG or O-RU (or RU).

[0444] A structured antenna architecture is expected in the rest of the disclosure. For simplicity, each PG or O-RU (or RU) is equivalent to a panel (cf. FIG. 13), although, an PG or O-RU (or RU) can have multiple panels in practice. The disclosure however is not restrictive to a single panel assumption at each PG or O-RU (or RU), and can easily be extended (covers) the case when an PG or O-RU (or RU) has multiple antenna panels.

[0445] In one embodiment, an PG (or O-RU or RU) constitutes (or corresponds to or is equivalent to) at least one of the following:

[0446] In one example, an PG or O-RU (or RU) corresponds to a TRP.

[0447] In one example, an PG or O-RU (or RU) corresponds to a DL RS (e.g. CSI-RS) or UL RS (e.g. SRS). A UE is configured with $K=N_{\text{sub.g}}>1$ DL or UL RSs, one per PG.

[0448] In one example, an PG or O-RU (or RU) corresponds to a DL/UL RS group, where a group comprises one or multiple DL/UL RSs. A UE is configured with $K \geq N_{\text{sub.g}}>1$ DL/UL RSs, and the K RSs can be partitioned into $N_{\text{sub.g}}$ groups. The information about the grouping can be provided together with the RS setting/configuration, or with the reporting setting/configuration.

[0449] In one example, an PG or O-RU (or RU) corresponds to a subset (or a group) of DL/UL RS ports.

[0450] In one

example, an PG or O-RU (or RU) corresponds to one or more examples described herein depending on a configuration. For example, this configuration can be explicit via a parameter (e.g. an RRC parameter). Or it can be implicit. [0451] In one example, when implicit, it could be based on the value of K. [0452] In another example, the configuration could be based on the configured codebook. For example, an PG or O-RU (or RU) corresponds to a CSI-RS resource (according to one or more examples described herein) or resource group (according to one or more examples described herein) when the codebook corresponds to a decoupled codebook (modular or separate codebook for each PG or O-RU (or RU)), and an PG or O-RU (or RU) corresponds to a subset (or a group) of CSI-RS ports (according to one or more examples described herein) when codebook corresponds to a coupled (joint or coherent) codebook (one joint codebook across PGs).

[0453] In one example, when PG or O-RU (or RU) maps (or corresponds to) a CSI-RS or CSI-RSs (according to one or more examples described herein), and a UE can select a subset of PGs (resources or resource groups) and report the CSI for the selected PGs (resources or resource groups), the selected PGs can be reported via an indicator. For example, the indicator can be a PGI (PG indicator) or CRI or SRI or a PMI (component) or a new indicator.

[0454] In one example, when PG or O-RU (or RU) maps (or corresponds to) a CSI-RS port group (according to one or more examples described herein), and a UE can select a subset of PGs (port groups) and report the CSI for the selected PGs (port groups), the selected PGs can be reported via an indicator. For example, the indicator can be a PGI or a CRI or a PMI (component) or a new indicator.

[0455] In one example, when multiple ($K > 1$) CSI-RSs are configured for N.sub.g PGs (according to one or more examples described herein), a decoupled (modular) codebook is used/configured, and when a single ($K = 1$) CSI-RS for N.sub.g PGs (according to one or more examples described herein), a joint codebook is used/configured.

[0456] In one embodiment, a UE is configured (e.g. via a higher layer) with a report (e.g. CSI report), where the report is based on a channel measurement (or, optionally, interference measurement or/and a codebook). When the report is configured to be aperiodic, it is reported when triggered via a DCI field (e.g. a CSI request field) in a DCI. The DCI can be UL-related DCI (DCI format with UL grant for PUSCH) or DL-related DCI (DCI format with DL assignment for PDSCH) or a dedicated special-purpose DCI.

[0457] The channel measurement can be based on $K \geq 1$ channel measurement RSs (CMRs or ports or resources) that are transmitted from a plurality of spatial-domain (SD) units (e.g. a SD unit = an antenna port), and are measured via a plurality of frequency-domain (FD) units (e.g. a FD unit = one or more PRBs/SBs) and via either a time-domain (TD) unit or a plurality of TD units (e.g. a TD unit = one or more time slots). In one example, a CMR can be a NZP-CSI-RS (or ports or resource) for DL channel measurement and a SRS (or ports or resource) for UL channel measurement. In case of DL-UL reciprocity, DL channel measurement can be based on UL channel measurement via SRS, and vice versa.

[0458] The report can be associated with the plurality of FD units and the plurality of TD units associated with the channel measurement. Alternatively, the report can be associated with a second set of FD units (different from the plurality of FD units associated with the channel measurement) or/and a second set of TD units (different from the plurality of TD units associated with the channel measurement). In this later case, the receiving entity (e.g. UE), based on the channel measurement, can perform prediction (interpolation or extrapolation) in the second set of FD units or/and the second set of TD units associated with the report.

[0459] An illustration of the SD units (in 1.sup.st and 2.sup.nd antenna dimensions), FD units, and, and TD units is shown in FIG. 14. [0460] The first dimension is associated with the 1st antenna port dimension and comprises N.sub.1 units, [0461] The second dimension is associated with the 2nd antenna port dimension and comprises N.sub.2 units, [0462] The third dimension is associated with the frequency dimension and comprises N.sub.3 units, and [0463] The fourth dimension is

associated with the time/Doppler dimension and comprises $N_{\text{sub.4}}$ units.

[0464] Alternatively, the SD units, FD units, and, and TD units are as follows. [0465] The first dimension is associated with the antenna port dimension and comprises $P_{\text{sub.tot}}$ units, [0466] The second dimension is associated with the frequency dimension and comprises $N_{\text{sub.3}}$ units, and [0467] The third dimension is associated with the time/Doppler dimension and comprises $N_{\text{sub.4}}$ units.

[0468] The plurality of SD units can be associated with antenna ports (e.g. co-located at one site or distributed across multiple sites) comprising one or multiple antenna/port groups (i.e., $N_{\text{sub.g}} \geq 1$), and dimensionalizes the spatial-domain profile of the channel measurement.

[0469] When $K=1$, there is one CMR comprising $P_{\text{sub.tot}}$ antenna ports. [0470] When $N_{\text{sub.g}}=1$, there is one PG comprising $P_{\text{sub.tot}}$ ports, and the report is based on the channel measurement from the one PG. [0471] When $N_{\text{sub.g}} > 1$, there are multiple PGs, and the report is based on the channel measurement from/across the multiple PGs.

[0472] When $K > 1$, there are multiple CMRs, and the report is based on the channel measurement across the multiple CMRs. In one example, a CMR corresponds to an PG (one-to-one mapping). In one example, multiple CMRs can correspond to an PG (many-to-one mapping).

[0473] In one example, when the $P_{\text{sub.tot}}$ antenna ports are co-located at one site, $N_{\text{sub.g}}=1$. In one example, when the $P_{\text{sub.tot}}$ antenna ports are distributed (non-co-located) across multiple sites, $N_{\text{sub.g}} > 1$.

[0474] In one example, when the $P_{\text{sub.tot}}$ antenna ports are co-located at one site and within a single antenna panel, $N_{\text{sub.g}}=1$. In one example, when the $P_{\text{sub.CSIRS}}$ antenna ports are distributed across multiple antenna panels (can be co-located or non-co-located), $N_{\text{sub.g}} > 1$.

[0475] The value of $N_{\text{sub.g}}$ can be configured, e.g. via higher layer RRC parameter. Or it can be indicated via a MAC CE. Or it can be provided via a DCI field.

[0476] Likewise, the value of K can be configured, e.g. via higher layer RRC parameter. Or it can be indicated via a MAC CE. Or it can be provided via a DCI field.

[0477] In one example, $K=N_{\text{sub.g}}=X$. The value of X can be configured, e.g. via higher layer RRC parameter. Or it can be indicated via a MAC CE. Or it can be provided via a DCI field.

[0478] In one example, the value of K is determined based on the value of $N_{\text{sub.g}}$. In one example, the value of $N_{\text{sub.g}}$ is determined based on the value of K .

[0479] The plurality of FD units can be associated with a frequency domain allocation of resources (e.g. one or multiple CSI reporting bands, each comprising multiple PRBs) and dimensionalizes the frequency (or delay)-domain profile of the channel measurement.

[0480] The plurality of TD units can be associated with a time domain allocation of resources (e.g. one or multiple CSI reporting windows, each comprising multiple time slots) and dimensionalizes the time (or Doppler)-domain profile of the channel measurement.

[0481] In one example, the number of antenna ports across K CMRs is the same. For example, each of the K CMRs can be associated with $2N_{\text{sub.1}}N_{\text{sub.2}}$ antenna ports. In this case, the total number of antenna ports is $P_{\text{sub.tot}, K=2KN_{\text{sub.1}}N_{\text{sub.2}}}$.

[0482] In one example, the number of antenna ports across K CMRs can be the same or different. For example, each of the K CMRs can be associated with $2N_{\text{sub.1}, r}N_{\text{sub.2}, r}$ antenna ports. In this case, the total number of antenna ports is $P_{\text{sub.tot}, K=\sum_{r=1}^{\text{sup}} K 2N_{\text{sub.1}, r}N_{\text{sub.2}, r}}$.

[0483] In port numbering scheme 1, the CMR ports are numbered according to the order of (polarization p , CMR r) as CMR ports of $(p=0, r=1)$ followed by CMR ports of $(p=1, r=1)$, followed by CMR ports of $(p=0, r=2)$, followed by CMR ports of $(p=1, r=2)$, . . . , followed by CMR ports of $(p=0, r=N)$ followed by CMR ports of $(p=1, r=N)$.

[0484] In port numbering scheme 2, the CMR ports are numbered according to the order of (polarization p , CMR r) as [0485] CMR ports of $(p=0, r=1)$ followed by CMR ports of $(p=0, r=1)$, . . . , followed by CMR ports of $(p=0, r=N)$, and [0486] then CMR ports of $(p=1, r=1)$ followed by CMR ports of $(p=1, r=1)$, . . . , followed by CMR ports of $(p=1, r=N)$.

[0487] In one example, PGs can have a uniform (the same/common) structure. For example, they can have the same number of ports ($P_{\text{sub.tot},r}=P_{\text{sub.tot}}$) or the same antenna port layout ($N_{\text{sub.1},r}, N_{\text{sub.2},r}=(N_{\text{sub.1}}, N_{\text{sub.2}})$). In one example, PGs can have non-uniform (or different) structure. For example, they can have the same or different number of ports ($P_{\text{sub.tot},r,\text{sub.1}}=P_{\text{sub.tot},r,\text{sub.2}}$ or $P_{\text{sub.tot},r,\text{sub.1}}\neq P_{\text{sub.tot},r,\text{sub.2}}$) or the same antenna port layout, i.e., ($N_{\text{sub.1},r,\text{sub.1}}, N_{\text{sub.2},r,\text{sub.1}}=(N_{\text{sub.1},r,\text{sub.2}}, N_{\text{sub.2},r,\text{sub.2}})$ or ($N_{\text{sub.1},r,\text{sub.1}}, N_{\text{sub.2},r,\text{sub.1}}\neq(N_{\text{sub.1},r,\text{sub.2}}, N_{\text{sub.2},r,\text{sub.2}})$).

[0488] In one embodiment, a user is configured with a MIMO operation (such as CSI reporting) based on two steps: [0489] 1) NW configuring a UE to measure DL RS(s) associated with $N_{\text{sub.g}}\geq 1$ ports or PG(s) [0490] 2) A reporting based on the measurement. For example, CSI reporting based on at least one CMR (and at least one IMR), and optionally, based on a codebook (e.g. for PMI).

[0491] In one example, properties/assumptions associated with port(s) of the one or multiple PGs can be established and indicated/configured to the UE. For instance, at least one QCL relation/info (such as TypeA/B/C/D in 5G NR) including one or more long-term (LT) channel properties (such as Doppler shift, Doppler spread, delay spread, average delay, spatial filter/relation) between port(s) can be configured. The QCL info(s) can be configured via a TCI state definition. A measurement and reporting to enable TCI state indication can be supported. Such reporting can be aperiodic: either NW-controlled (via DCI either UL-DCI or DL-DCI or a dedicated DCI, including two-stage DCI) or UE-initiated (e.g. UCI either PUCCH or PUSCH or dedicated layer 1 UL channel, including two-stage UCI). The UE-initiated can be based on an event detection.

[0492] Akin to NR UL codebook, the Coherency between/within PGs can be (1) full-coherence (FC), i.e., a layer (stream) is transmitted using antenna ports, (2) partial-coherence (PC), i.e., a layer (stream) is transmitted using at least two but not each of the antenna ports, or (3) non-coherence (NC), i.e., a layer (stream) is transmitted using one antenna port.

[0493] In one embodiment, a UE (e.g., the UE 116) is configured (e.g. via higher layer) with a measurement procedure based on at least one of the following components (or information). [0494] Component (A): measurement resources/region (referred to as T-F-resource-info), which is a set of time-frequency (T-F) resources (e.g. REs) S. In one example, S is a pattern (from multiple patterns) of REs in a PRB or across multiple PRBs (e.g. 2). The component A also includes the frequency band (PRBs for measurement) and a value of periodicity/offset when the measurement is periodic. [0495] Component (B): port-related information (referred to as port-info), such as a number of ports $N_{\text{sub.port}}$, density (ρ) of each ports (e.g. in terms of number of REs/port/PRB), and CDM-Type (if used). [0496] Component (C): RS-related information (referred to as RS-info), such as RS-type=zero-power (ZP) or non-zero-power (NZP), and when RS-type=NZP, information such as power control offset and scrambling. In one example, the component C can also include additional information such as QCL-info (or TCI state)

[0497] A summary of the three components and an example of each is provided in Table 3.

TABLE-US-00012 TABLE 3 Component Information Example A: T-F- T-F REs S

ResourceMapping resource- frequencyDomainAllocation info firstOFDMsymbolInTimeDomain
ResourceElementPattern pattern0, pattern1, . . . freqBand FrequencyOccupation set of RBs (e.g.
startingRB, nrofRBs) periodicityAndOffset B: port- Port ResourceMapping info nrofPorts, density,
cdm-Type C: RS-info Power powerControlOffset, scramblingID powerControlOffset2 QCL-info or
TCI state AdditionalInfo

[0498] In one example, the resourceElementPattern (in the table herein) is one of pattern0 and pattern1 as in Table 4. In one example, for RS-Type=CSI-IM, only one pattern from resourceElementPattern can be configured.

TABLE-US-00013 TABLE 4 ResourceElementPattern CHOICE { pattern0 SEQUENCE {
subcarrierLocation-p0 ENUMERATED { s0, s2, s4, s6, s8, s10 },
symbolLocation-p0 INTEGER (0..12) }, pattern1 SEQUENCE {

subcarrierLocation-p1 ENUMERATED { s0, s4, s8 }, symbolLocation-p1

INTEGER (0..13) }

[0499] When $p=0.5$, a port is mapped to (hence can be measured via) an RE in even-numbered or odd-numbered PRBs, when $p=1$, a port is mapped to (hence can be measured via) via an RE in each PRB; when $p=3$, a port is mapped to (hence can be measured via) 3 REs in each PRB, and so on.

[0500] Likewise, when $N.sub.port > 1$, each port $p.sub.i$ is assigned a subset of REs $S.sub.i$, where $i \in \{0, \dots, N.sub.port-1\}$ or $\in \{1, \dots, N.sub.port\}$, where the value $p.sub.i = X+i$ and $X = \text{port number of the first of the } N.sub.port \text{ ports}$.

[0501] In one example, for a measurement region (component A), a measurement entity can be configured according to at least one of the following examples. [0502] In one example, the

measurement entity includes $N.sub.port \geq 1$ port(s) or PGs or a set of ports/PGs. [0503] In one

example, the measurement entity includes an RS or a set of RSs (note: an RS here essentially has one-port). [0504] In one example, the measurement entity includes a pair (RS r , $N.sub.port, r$) or a

set of pairs, where RS r is associated with $N.sub.port, r$ ports.

[0505] In one example, a mapping (association) of the measurement entity to the T-F measurement

REs can be based on [0506] In one example, the mapping is based on T-F patterns. [0507] In one

example, the mapping is based on a bitmap.

[0508] In one example, a set of ports can be configured (via RRC) or indicated (via MAC CE or DCI). The indication can be via a bitmap or combinatorial indicator, which indicates $N.sub.port$ out of M ports.

[0509] In one example, the set of RSs $r=1, \dots, N.sub.RS$ can be indicated via a value or ID of each RS r .

[0510] In one example, the T-F REs is configured via a parameter or an IE (e.g. ResourceMapping).

[0511] In one example, the parameter/IE includes at least two parameters for FD and TD

allocations. In one example, the two parameters are frequencyDomainAllocation and

firstOFDMSymbolInTimeDomain. An example is shown in Table 5. [0512] In one example, the

parameter/IE includes one parameter for indication of FD and TD allocations jointly.

TABLE-US-00014 TABLE 5 frequencyDomainAllocation CHOICE { row1 BIT

STRING (SIZE (4)), row2 BIT STRING (SIZE (12)), row4 BIT STRING

(SIZE (3)), other BIT STRING (SIZE (6)) }, firstOFDMSymbolInTimeDomain

INTEGER (0..13) or (2..12)

[0513] In one example, the port-info is configured via a parameter/IE (e.g. PortMapping). [0514] In

one example, the parameter/IE includes at least three parameters for $N.sub.port$, p and CDM-Type.

In one example, the three parameters are as shown in Table 6. [0515] In one example, the

parameter/IE includes one parameter for indication of $N.sub.port$, p and CDM-Type jointly.

TABLE-US-00015 TABLE 6 nrofPorts ENUMERATED {p1,p2,p4,p8,p12,p16,p24,p32}, cdm-

Type ENUMERATED {noCDM, fd-CDM2, cdm4-FD2-TD2, cdm8-FD2- TD4}, density

CHOICE { dot5 ENUMERATED {evenPRBs, oddPRBs}, one NULL,

three NULL, spare NULL },

[0516] In one example, the frequency band is configured via a parameter/IE (e.g. freqBand). [0517]

In one example, the parameter/IE is as shown in Table 7.

TABLE-US-00016 TABLE 7 freqBand CSI-FrequencyOccupation, CSI-FrequencyOccupation ::=

SEQUENCE { startingRB INTEGER (0..maxNrofPhysicalResourceBlocks-1),

nrofRBs INTEGER (24..maxNrofPhysicalResourceBlocksPlus1), ... }

[0518] In one example, there is a restriction on the granularity of the frequency band. For example, the restriction can correspond to that number of RBs being a multiple of a number (e.g. 4) or/and a minimum value of the frequency band or BWP size (e.g. 24).

[0519] In one embodiment, a UE is configured (via higher layer parameter(s) or IE(s)) with the component (A) for measurement. The UE is further provided with an information about a RS-type

(e.g. included in component C), where the RS-type takes a value from a set T. [0520] In one example, T includes {NZP, ZP}. Two examples are shown in Table 8 and Table 9. [0521] In one example, T includes {NZP, CSI-IM}. [0522] In one example, T includes {ZP, CSI-IM}. [0523] In one example, T includes {NZP, ZP, CSI-IM}. An example is shown in Table 10.

[0524] The information about a RS-type can be provided a DCI. [0525] In one example, the DCI can be a UL-related DCI. [0526] In one example, the DCI can be a DL-related DCI. [0527] In one example, the DCI can be a dedicated (special purpose) DCI. [0528] In one example, the DCI can be a DL-related or UL-related DCI. For example, when RS-Type=ZP, the DCI is a DL-related DCI, and when RS-Type=NZP or CSI-IM, the DCI is UL-related DCI. [0529] In one example, the DCI (one of the mentioned herein) can be a two-stage DCI (DCI1, DCI2), and the information about the RS-Type can be provided via DCI1. [0530] In one example, the DCI (one of the mentioned herein) can be a two-stage DCI (DCI1, DCI2), and the information about the RS-Type can be provided via DCI2. [0531] In one example, the DCI (one of the mentioned herein) can be a two-stage DCI (DCI1, DCI2), and the information about the RS-Type can be provided via DCI1 or/and DCI2.

[0532] The information about a RS-type can be provided a MAC CE. [0533] In one example, the MAC CE can be a UL-related MAC CE. [0534] In one example, the MAC CE can be a DL-related MAC CE. [0535] In one example, the MAC CE can be a dedicated (special purpose) MAC CE. [0536] In one example, the MAC CE can be a DL-related or UL-related MAC CE. For example, when RS-Type=ZP, the MAC CE is a DL-related MAC CE, and when RS-Type=NZP or CSI-IM, the MAC CE is UL-related MAC CE. [0537] In one example, the MAC CE (one of the mentioned herein) can be a two-stage MAC CE (MAC CE1, MAC CE2), and the information about the RS-Type can be provided via MAC CE1. [0538] In one example, the MAC CE (one of the mentioned herein) can be a two-stage MAC CE (MAC CE1, MAC CE2), and the information about the RS-Type can be provided via MAC CE2. [0539] In one example, the MAC CE (one of the mentioned herein) can be a two-stage MAC CE (MAC CE1, MAC CE2), and the information about the RS-Type can be provided via MAC CE1 or/and MAC CE2.

[0540] The information about a RS-type can be provided a combination of MAC CE and DCI, where the MAC CE or/and DCI is according to one of the examples mentioned herein.

[0541] When the RS-Type=NZP, the UE is configured to measure a channel based on the NZP RS (e.g. CSI-RS). [0542] In one example, the channel can be a desired channel (i.e., the NZP RS is a CMR). [0543] In one example, the channel can be an interference channel (i.e., the NZP RS is an IMR). [0544] In one example, the NZP RS can be a CMR or IMR based on RRC or/and MAC CE or/and DCI based signalling.

[0545] When the RS-Type=CSI-IM, the UE is configured to measure an interference channel based on the T-F REs configured via the component (A) (i.e., the CSI-IM is an EVIR).

[0546] When the RS-Type=ZP, [0547] In one example, the UE is not expected to measure the T-F REs configured via the component (A). In one example, such RS-Type can only be configured together with a PDSCH-Config. [0548] In one example, the UE can be configured to measure an interference channel based on the T-F REs configured via the component (A) (i.e., the ZP RS is an IMR). [0549] In one example, the ZP RS can for one of the two examples based on RRC or/and MAC CE or/and DCI based signalling.

[0550] When the RS-Type=ZP, [0551] In one example, the corresponding information corresponds to those in the component (A) only. That is, the ZP RS does not require any port-related information. [0552] In one example, the corresponding information corresponds to those in the component (A) and (B). That is, the ZP RS requires port-related information.

[0553] In one example, there is no separate configuration for CSI-IM, and the RS-Type=CSI-IM is a special configuration/case of ZP. For example, ZP RS for interference (ZP IMR). [0554] In one example, the UE can indicate via MAC CE or/and DCI with an information indicating that the configured ZP RS is to be used for interference measurement. This information can be included in the UL-DCI that triggers AP CSI report. [0555] In one example, ZP RS can correspond to two sets

of T-F REs (or patterns), S1 and S2, where the set S2 can corresponds to the case when ZP RS=CSI-IM.

TABLE-US-00017 TABLE 8 NZP ZP T-F pattern CSI-RS-ResourceMapping A

frequencyDomainAllocation = {Row1, row2, row4, other} firstOFDMSymbolInTimeDomain {(0...13), (2...12)} csi-IM-ResourceElementPattern These pattern0 = (2, 2) patterns are pattern1 = (4, 1) included in ZP/NZP patterns freqBand CSI-FrequencyOccupation Multiple of startingRB 4 and min nrofRBs #RBs = 24 periodicityAndOffset Port CSI-RS-ResourceMapping B nrofPorts = {1, 2, 4, 8, 12, 16, 24, 32} Density = {0.5, 1, 3} cdm-Type = {noCDM, fd-CDM2, cdm4-FD2-TD2, cdm8-FD2-TD4} Power powerControlOffset = (-8...15) C powerControlOffsetSS = {db-3, db0, db3, db6} scramblingID QCL-info for P CSI-RS AdditionalInfo LTM related

TABLE-US-00018 TABLE 9 NZP ZP T-F pattern CSI-RS-ResourceMapping A

frequencyDomainAllocation = {Row1, row2, row4, other} firstOFDMSymbolInTimeDomain {(0...13), (2...12)} csi-IM-ResourceElementPattern These pattern0 = (2, 2) patterns are pattern1 = (4, 1) included in ZP/NZP patterns freqBand CSI-FrequencyOccupation Multiple of startingRB 4 and min nrofRBs #RBs = 24 periodicityAndOffset Port CSI-RS-ResourceMapping B nrofPorts = {1, 2, 4, 8, 12, 16, 24, 32} Density = {0.5, 1, 3} cdm-Type = {noCDM, fd-CDM2, cdm4-FD2-TD2, cdm8-FD2-TD4} Power powerControlOffset = (-8...15) C powerControlOffsetSS = {db-3, db0, db3, db6} scramblingID QCL-info for P CSI-RS AdditionalInfo LTM related

TABLE-US-00019 TABLE 10 NZP ZP CSI-IM T-F pattern CSI-RS-ResourceMapping A

frequencyDomainAllocation = {Row1, row2, row4, other} firstOFDMSymbolInTimeDomain {(0..13), (2..12)} csi-IM-ResourceElementPattern These pattern0 = (2, 2) patterns are pattern1 = (4, 1) included in ZP/NZP patterns freqBand CSI-FrequencyOccupation Multiple of startingRB 4 and min nrofRBs #RBs = 24 periodicityAndOffset Port CSI-RS-ResourceMapping B nrofPorts = {1, 2, 4, 8, 12, 16, 24, 32} Density = {0.5, 1, 3} cdm-Type = {noCDM, fd-CDM2, cdm4-FD2-TD2, cdm8-FD2-TD4} Power powerControlOffset = (-8..15) C powerControlOffsetSS = {db-3, db0, db3, db6} scramblingID QCL-info for P CSI-RS AdditionalInfo LTM related

[0556] In one example, the RS can be a CSI-RS and RS-Type can be NZP or ZP. [0557] When RS-Type=NZP, the CSI-RS is used as a CMR (or IMR if configured for interference measurement). [0558] When RS-Type=ZP, the CSI-RS is either (i) used as an IMR or (ii) not used for PDSCH reception (or rate matching). For instance, it is (i) when the CSI-RS is linked with a CSI report, and it is (ii) when the CSI-RS is linked with a PDSCH-Config.

[0559] In one example, when RS=CSI-RS and the RS-Type=NZP, the relevant information corresponds to three components (A), (B) and (C). In one example, when RS=CSI-RS and the RS-Type=ZP, the relevant information corresponds to the component (A) alone or components (A) and (B).

[0560] In one example, a UE is configured with at least one CMR and at least one IMR that are linked with a CSI report. The UE measures channel using the at least one CMR and interference using the least one IMR and determines the CSI report based on the measurement. The CMR and IMR resources are CSI-RSs, when CSI-RSs are CMRs, then NZP related info (components B and C) are also provided for this CSI-RSs.

[0561] In one example, a UE is configured with a measurement resource, i.e., the component (A), which can map to $T \geq 1$ different configuration for the component (B) or (port-related info), i.e., one (A) can map to multiple (B)s. In one example, the component (A) can be signaled via RRC, and one of the T different configurations for (B) can be indicated via DCI or/and MAC CE.

[0562] In one example, a UE (e.g., the UE 116) is configured with each of or a subset of information in the component (C) only when the RS-Type=NZP. [0563] In one example, the component (C) is signaled together with the component (B). For example, both via DCI or (B) via MAC CE and (C) via DCI. [0564] In one example, the component (C) is signaled separately from

the component (B). [0565] In one example, the component (C) is signaled via DCI.

[0566] In one example, the subset of information corresponds to power-related information. In one example, the power-related info can be indicated via DCI (e.g. for NES, CLI measurement for duplex).

[0567] In one example, ZP CSI-RS can be UE-specific since it is included in PDSCH-Config.

[0568] For one UE, an RS can be ZP and can be used as blanking or rate matching, and for another UE, it can be NZP. [0569] In one example, for UE1, ZP associated with (A), and for UE2, NZP associated with (A) and (B), where (A) for one UE can be the same or included in that for other UE. [0570] In one example, if the UE is configured with (A) and (B), the corresponding RS-Type has to be NZP, i.e., it can't be ZP (or CSI-IM). [0571] In one example, if the UE is configured with (A) and (B), the corresponding RS-Type can be any of NZP, ZP, CSI-IM. For example, the power offset value can be used to indicate the muting of RS-info. For example, (B) can be muted via lower layer (DCI), hence it can be used as ZP. [0572] If A is RRC configured, then it can be used as both ZP and NZP.

[0573] In one example, ZP CSI-RS can be pre-coded (beamformed) or non-precoded. In one example, CSI-IM is UE-common, and non-precoded

[0574] In one example, the location of ZP CSI RS is tightly associated with the PRB scheduled for PDSCH reception. In one example, the location of ZP CSI RS can be outside the scheduled PDSCH.

[0575] In one example, when RS-Type=NZP, the RS is CSI-RS and the purpose (use case) of the measurement is to facilitate at least one of the following. [0576] In one example, the NZP RS is for CSI report. [0577] In one example, the NZP RS is for tracking. [0578] In one example, the NZP RS is for mobility. [0579] In one example, the NZP RS is for beam report (L1-RSRP/L1-SINR). [0580] In one example, the NZP RS is for time-domain channel property (TDCP) report. [0581] In one example, the NZP RS is for CJT calibration report (time/delay offset, frequency offset, or/and phase offset) associated with $N_{\text{sub}} > 1$ NZP RSs. [0582] In one example, the NZP RS is for positioning (PRS). [0583] In one example, the NZP RS is for phase tracking RS (PTRS). [0584] In one example, the NZP RS is for pathloss RS (PL RS).

[0585] In one example, a UE is configured (RRC) with one measurement region (component A), which can include (map to) at least one CSI-RS configuration, except that RS-Type=NZP or ZP is not configured via higher layer (RRC) and is rather indicated via MAC CE or/and DCI.

[0586] In one example, the (RS-Type=) ZP CSI-RS is triggered via DL-DCI, and requires less RRC-configured parameters (only component A) than (RS-Type=) NZP CSI-RS.

[0587] When the DCI is a dedicated DCI (neither DL-DCI nor UL-DCI), a trigger can be according to: [0588] In Ex1, one DCI field (or code point) is used to indicate ZP or NZP. The field can also be a ZP trigger (for PDSCH reception) or a NZP trigger (for AP CSI); hence the ZP trigger or the CSI trigger in DL-DCI or UL-DCI, respectively, can be absent (not needed). Or, if present, they can be ignored, or they can override the respective values. [0589] In Ex2, two DCI fields (or code points) are used, ZP field and a NZP field. The ZP field can also be a ZP trigger (for PDSCH reception). The NZP field can also be a NZP trigger (for CSI trigger).

[0590] When DCI=DL-DCI (a DL trigger for PDSCH reception), [0591] In Ex1, a ZP trigger field (as in 5G NR) is used/indicated. For example, a up to 2-bit field is used where Field value 00 is reserved. [0592] In Ex2, a ZP trigger field (as in 5G NR) is used but ZP or NZP can also be indicated. In one example, one field value (e.g. 00) can be used to indicate NZP, and remaining values (e.g. 01, 10, 11) can be used to indicate ZP. When NZP is indicated, [0593] In one example, it can just switch/assign the role (or use case) of the CSI-RS (ZP to NZP to ZP and so on). [0594] In one example, it can also indicate/trigger a AP NZP CSI-RS measurement, and may also accompany a CSI report. [0595] In Ex3, a field (or code point) is used/indicated to dynamically switch/assign ZP or NZP.

[0596] When DCI=UL-DCI (a UL trigger for PUSCH grant) [0597] In Ex1, a NZP trigger field,

similar to ZP trigger field is used. [0598] In Ex2, a CSI trigger field, it is NZP. The additional parameters of NZP can be configured via MAC CE or/and DCI.

[0599] In one example, the UE can be configured with a list of CSI-RSs that can map to (configured in) the configured measurement region (component A). [0600] In one example, the trigger (explained herein) is per CSI-RS. In one example, the trigger can be a bitmap with one bit associated with each CSI-RS. [0601] In one example, the trigger (explained herein) is a common trigger (for CSI-RSs).

[0602] In one example, an IE to configure a CSI-RS is given by Table 11.

TABLE-US-00020 TABLE 11 CSI-RS ::= SEQUENCE { csi-RS-Id CSI-RS-Id, resourceMapping CSI-RS-ResourceMapping, periodicityAndOffset CSI-ResourcePeriodicityAndOffset OPTIONAL, -- Cond PeriodicOrSemiPersistent }

[0603] In one example, information about NZP CSI-RS (e.g. triggered via DCI (e.g. UL-DCI)) is given by Table 12. There can be additional parameters, e.g. as in Table 13.

TABLE-US-00021 TABLE 12 NZP-CSI-RS-Param ::= SEQUENCE { powerControlOffset INTEGER (-8..15), powerControlOffsetSS ENUMERATED{db-3, db0, db3, db6} OPTIONAL, -- Need scramblingID ScramblingId, additionalParams AdditionalParams }

TABLE-US-00022 TABLE 13 AdditionalParams ::= SEQUENCE { RS TCI-StateId OPTIONAL, -- Cond Periodic ltmParams LTMPParams OPTIONAL, -- Cond LTM } LTMPParams ::= SEQUENCE { subcarrierSpacing-r18 SubcarrierSpacing OPTIONAL, absoluteFrequencyPointA-r18 ARFCN-ValueNR OPTIONAL, cyclicPrefix-r18 ENUMERATED {extended} OPTIONAL }

[0604] In one example, a UE is configured with the components (A, B, C) according to at least one of the following examples. [0605] In one example, (A, B, C) is signalled via (RRC, RRC, RRC) either together in one IE or separately in 2 or 3 IEs. [0606] In one example, (A, B, C) is signalled via (RRC, RRC, DCI) and (A, B) either together in one IE or separately in 2 IEs. The example of the DCI is according to one of the examples of DCI mentioned herein. [0607] In one example, (A, B, C) is signalled via (RRC, DCI, RRC) and (A, C) either together in one IE or separately in 2 IEs. The example of the DCI is according to one of the examples of DCI mentioned herein. [0608] In one example, (A, B, C) is signalled via (RRC, RRC, MAC CE) and (A, B) either together in one IE or separately in 2 IEs. The example of the MAC CE is according to one of the examples of MAC CE mentioned herein. [0609] In one example, (A, B, C) is signalled via (RRC, MAC CE, RRC) and (A, C) either together in one IE or separately in 2 IEs. The example of the MAC CE is according to one of the examples of MAC CE mentioned herein. [0610] In one example, (A, B, C) is signalled via (RRC, MAC CE, DCI). The example of the DCI or MAC CE is according to one of the examples of DCI or MAC CE mentioned herein. [0611] In one example, (A, B, C) is signalled via (RRC, DCI, MAC CE). The example of the DCI or MAC CE is according to one of the examples of DCI or MAC CE mentioned herein. [0612] In one example, (A, B, C) is signalled via (RRC, MAC CE1, MAC CE2). The MAC CE1 and MAC CE2 are two MAC CEs, or two parts of a two-stage MAC CE as mentioned herein. [0613] In one example, (A, B, C) is signalled via (RRC, DCI1, DCI2). The DCI1 and DCI2 are two DCI1 or two parts of a two-stage DCI as mentioned herein.

[0614] In one example, a set of ports or a PG can be mapped to (or determine based on) a T-F resource subset/partition of the T-F measurement region (component A). In one example, the T-F resource (e.g. pattern)=T or its subsets T/r where r is an integer, and is equal to or a multiple of a value of N.sub.port. [0615] In one example, a value of number of ports can be fixed and a multiple of 4, e.g. 32, 24, 16, 12, 8 etc. [0616] In one example, a value of number of ports can be fixed and a multiple 2 or 4, e.g. 32, 24, 16, 12, 10, 8, 6, 4 etc. [0617] In one example, a value of number of ports can be fixed and a multiple of 4 or a few odd values, e.g. 32, 24, 16, 12, 8, 5, 4, 3, 2, 1 etc. [0618] In one example, a value of number of ports can be any integer value

[0619] In one example, a set of ports or a PG can be mapped to (or determine based on) a T-F resource which is a superset/aggregation of multiple sets. In one example, T-F resource (e.g. pattern)=T or its supersets $T \times s$ where s is an integer and is equal to or a multiple of a value of $N_{\text{sub.port}}$.

[0620] FIG. 21 illustrates an example method 2100 performed by a UE in a wireless communication system according to embodiments of the present disclosure. The method 2100 of FIG. 21 can be performed by any of the UEs 111-116 of FIG. 1, such as the UE 116 of FIG. 3, and a corresponding method can be performed by any of the BSs 101-103 of FIG. 1, such as BS 102 of FIG. 2. The method 2100 is for illustration only and other embodiments can be used without departing from the scope of the present disclosure.

[0621] The method 2100 begins with the UE receiving information about a measurement procedure including a measurement resource including a set of T-F resource-info, port-info, and/or RS-info (2110). For example, in 2110, the T-F resource-info includes either TD and FD allocations or a T-F pattern and the port-info includes at least one of a number of ports, a density of each port, and a CDM type.

[0622] In various embodiments, the RS-info includes at least one RS-type from {ZP, NZP}, and when the at least one RS-type is NZP, further includes at least one of a power control offset and scrambling information. In one example, when the measurement procedure is configured as IM, the at least one RS-type is ZP and the information includes the T-F-resource-info and does not include the port-info. In another example, when the measurement procedure is configured as a CM, the at least one RS-type is NZP and the information includes the T-F-resource-info, the port-info and the RS-type info.

[0623] The UE then receives a DL signal according to the information (2120). In various embodiments, the UE may also receive a trigger via DCI that indicates at least one slot associated with the DL signal.

[0624] The UE then performs the measurement procedure based on the information and the received DL signal (2130). In various embodiments, the UE further receives an information about a CSI report associated with the measurement procedure, determines the CSI report, and transmits the report including at least one indicator indicating the CSI report.

[0625] Any of the above variation embodiments can be utilized independently or in combination with at least one other variation embodiment. The above flowcharts illustrate example methods that can be implemented in accordance with the principles of the present disclosure and various changes could be made to the methods illustrated in the flowcharts herein. For example, while shown as a series of steps, various steps in each figure could overlap, occur in parallel, occur in a different order, or occur multiple times. In another example, steps may be omitted or replaced by other steps.

[0626] Any of the above variation embodiments can be utilized independently or in combination with at least one other variation embodiment. The above flowchart(s) illustrate example methods that can be implemented in accordance with the principles of the present disclosure and various changes could be made to the methods illustrated in the flowcharts herein. For example, while shown as a series of steps, various steps in each figure could overlap, occur in parallel, occur in a different order, or occur multiple times. In another example, steps may be omitted or replaced by other steps.

[0627] Although the present disclosure has been described with exemplary embodiments, various changes and modifications may be suggested to one skilled in the art. It is intended that the present disclosure encompass such changes and modifications as fall within the scope of the appended claims. None of the descriptions in this application should be read as implying that any particular element, step, or function is an essential element that must be included in the claims scope. The scope of patented subject matter is defined by the claims.

Claims

1. A user equipment (UE) comprising: a transceiver configured to: receive information about a measurement procedure, the information including at least one of: at least one measurement resource including a set of time-frequency resource elements (REs) (T-F resource-info), port-related information (port-info), and reference signal (RS)-related information (RS-info), and receive a downlink (DL) signal according to the information; and a processor operably coupled to the transceiver, the processor configured to, based on the information and the received DL signal, perform the measurement procedure.
2. The UE of claim 1, wherein the T-F resource-info includes either time domain (TD) and frequency domain (FD) allocations or a T-F pattern.
3. The UE of claim 1, wherein the port-info includes at least one of a number of ports, a density of each port, and a code division multiplexing (CDM) type.
4. The UE of claim 1, wherein the RS-info includes at least one RS-type from {zero power (ZP), non-zero power (NZP)}, and when the at least one RS-type is NZP, further includes at least one of a power control offset and scrambling information.
5. The UE of claim 4, wherein, when the measurement procedure is configured as interference measurement (IM), the at least one RS-type is ZP and the information (i) includes the T-F-resource-info and (ii) does not include the port-info.
6. The UE of claim 4, wherein, when the measurement procedure is configured as a channel measurement (CM), the at least one RS-type is NZP and the information includes the T-F-resource-info, the port-info and the RS-type info.
7. The UE of claim 1, wherein the transceiver is further configured to receive a trigger via downlink control information (DCI) that indicates at least one slot associated with the DL signal.
8. The UE of claim 1, wherein: the transceiver is further configured to receive an information about a channel state information (CSI) report associated with the measurement procedure, the processor is further configured to determine the CSI report, and the transceiver is further configured to transmit the report including at least one indicator indicating the CSI report.
9. A base station (BS) comprising: a processor; and a transceiver operably coupled to the processor, the transceiver configured to: transmit information about a measurement procedure, the information including at least one of: at least one measurement resource including a set of time-frequency resource elements (REs) (T-F resource-info), port-related information (port-info), and reference signal (RS)-related information (RS-info), and transmit a downlink (DL) signal according to the information, wherein the measurement procedure is based on the information and the DL signal.
10. The BS of claim 9, wherein the T-F resource-info includes either time domain (TD) and frequency domain (FD) allocations or a T-F pattern.
11. The BS of claim 9, wherein the port-info includes at least one of a number of ports, a density of each port, and a code division multiplexing (CDM) type.
12. The BS of claim 9, wherein the RS-info includes at least one RS-type from {zero power (ZP), non-zero power (NZP)}, and when the at least one RS-type is NZP, further includes at least one of a power control offset and scrambling information.
13. The BS of claim 12, wherein, when the measurement procedure is configured as interference measurement (IM), the at least one RS-type is ZP and the information (i) includes the T-F-resource-info and (ii) does not include the port-info.
14. The BS of claim 12, wherein, when the measurement procedure is configured as a channel measurement (CM), the at least one RS-type is NZP and the information includes the T-F-resource-info, the port-info and the RS-type info.
15. The BS of claim 9, wherein the transceiver is further configured to transmit a trigger via downlink control information (DCI) that indicates at least one slot associated with the DL signal.

- 16.** The BS of claim 9, wherein the transceiver is further configured to: transmit an information about a channel state information (CSI) report associated with the measurement procedure, and receive the report including at least one indicator indicating the CSI report.
- 17.** A method performed by a user equipment (UE), the method comprising: receiving information about a measurement procedure, the information including at least one of: at least one measurement resource including a set of time-frequency resource elements (REs) (T-F resource-info), port-related information (port-info), and reference signal (RS)-related information (RS-info), and receiving a downlink (DL) signal according to the information; and based on the information and the received DL signal, performing the measurement procedure.
- 18.** The method of claim 17, wherein the T-F resource-info includes either time domain (TD) and frequency domain (FD) allocations or a T-F pattern.
- 19.** The method of claim 17, wherein the port-info includes at least one of a number of ports, a density of each port, and a code division multiplexing (CDM) type.
- 20.** The method of claim 17, wherein the RS-info includes at least one RS-type from {zero power (ZP), non-zero power (NZP)}, and when the at least one RS-type is NZP, further includes at least one of a power control offset and scrambling information.
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